
Information-Induced Training Geometry: Exact Reduction, Canonical Completion, and Structured Expressivity

Zavier Li
Xidian University
Xi'an, China

zavierli888@gmail.com

Abstract

Training data constrains optimizer geometry through the covectors visible to a declared information channel. We study how such partial information determines a full positive cometric relative to a reference and which degrees of freedom remain unidentified. Our central result resolves the full-column-rank SPD channel $\pi_A(P) = A^\top P A$ under affine-invariant Riemannian geometry. The map is a split-Hadamard metric submetry and admits an explicit unique completion $\Psi_{P_0, A}$, which is the AIRM-nearest full geometry realizing a visible target and yields exact full-to-visible variational reduction. When the channel moves, the completions form a gauge-invariant rank stratification of the SPD cone. Its closed-form pullback pair metric separates visible-metric motion from subspace rotation through the mismatch weight $R + R^{-1} - 2I$, yields an explicit positive-semidefinite multi-direction Gram matrix, and exposes the precise singularity of reference-valued rotation modes. The mechanism is explained by a metric theorem equating ball submetry, attained fiber distance, and lossless reduction of every monotone radial visible decision problem; a smooth split-Hadamard theorem supplies coherent information sheets, proximal commutation, and solution-wise gradient-flow lifting. The SPD realization also gives closed-form prior-data shrinkage. Diagonal and block optimizer families then reduce to relative-interior conic image tests with valid facial certificates, while deterministic and finite-sample bounds quantify recovery of the visible geometry and its subspace. Together these results characterize exact reduction, reference-dependent completion, and structured expressivity for the stated finite-dimensional AIRM model.

1 Introduction

Let $\mathbb{S}_{++}^n$ denote the cone of real symmetric positive definite matrices. A full-column-rank matrix $A \in \mathbb{R}^{n \times m}$ defines the positive compression

$$\pi_A : \mathbb{S}_{++}^n \longrightarrow \mathbb{S}_{++}^m, \quad \pi_A(P) = A^\top P A.$$

Given a reference $P_0 \in \mathbb{S}_{++}^n$ and a target $R \in \mathbb{S}_{++}^m$, the basic matrix problem studied here is

$$\min_{P \in \mathbb{S}_{++}^n} \{d_{\text{AI}}(P_0, P) : A^\top P A = R\}. \quad (1)$$

The constraint specifies only the compression of P , so its feasible set is a noncompact positive-definite fiber. We ask when this fiber has a unique affine-invariant nearest point, whether the constrained value reduces exactly to an m -dimensional problem, and how the selected completion changes when the matrix channel itself moves.

This compression problem is also the matrix core of partial-information training geometry. Adaptive optimization is often described as a choice of positive geometry: natural gradient uses the Fisher metric

(Amari, 1998); K-FAC and related methods use structured Fisher or Gauss–Newton approximations (Martens & Grosse, 2015; George et al., 2018); AdaGrad and Adam accumulate diagonal statistics (Duchi et al., 2011; Kingma & Ba, 2015); and Shampoo-type methods use matrix or tensor preconditioners (Gupta et al., 2018; Morwani et al., 2024). These methods differ in how they construct and restrict a positive map from covectors to parameter directions.

To make the information channel explicit, fix a parameter point θ , let $V = T_\theta \mathcal{M}$ and $E = V^*$, and write the per-example loss differentials as $a_i = d_\theta \ell_i \in E$. Their span

$$W = \text{span}\{a_1, \dots, a_n\} \subset E$$

is the first-order visible cotangent space. Its annihilator $\text{Ann}(W) \subset V$ is invisible to every example loss at that point. A declared positive visible target $R \in \mathbb{S}_{++}(W)$ constrains the restriction of a full cometric $P \in \mathbb{S}_{++}(E)$, but it leaves an entire fiber

$$\{P \in \mathbb{S}_{++}(E) : P|_{W \times W} = R\}$$

undetermined. The central questions are geometric and matrix-variational:

Which positive compressions admit lossless reduction and a canonical shortest completion, and how do channel motion and structural restrictions change the completed matrix geometry?

This question has three levels. At the first, no linearity or smoothness is available. We show that exact reduction of every reference-radial visible decision problem characterizes metric submetrics. This maximal statement isolates the precise property required for lossless completion.

At the second level, the spaces are Hadamard manifolds and the information map is a Riemannian submersion. If its horizontal distribution integrates to complete totally geodesic leaves, each leaf is a global isometric information sheet. The resulting section is coherent across updates, is the unique shortest lift, and commutes exactly with proximal maps and gradient flows.

At the third level, the ambient geometry is the SPD cone with the affine-invariant Riemannian metric (AIRM), and visible information is a full-column-rank linear channel

$$\pi_A(P) = A^\top P A.$$

This realization is fully explicit. For reference P_0 , set $R_0 = A^\top P_0 A$ and $U_0 = P_0^{1/2} A R_0^{-1/2}$. The canonical completion of $R \in \mathbb{S}_{++}^m$ is

$$\Psi_{P_0, A}(R) = P_0^{1/2} \left[I + U_0 (R_0^{-1/2} R R_0^{-1/2} - I) U_0^\top \right] P_0^{1/2}. \quad (2)$$

It is the unique AIRM-nearest point in the visible fiber, and the full deformation cost equals the visible cost exactly.

The explicit realization reveals additional structure. Modulo visible coordinate changes, active information descriptors (A, R) stratify the entire SPD cone by $\text{rank}(P - P_0)$. In normalized horizontal coordinates, the exact infinitesimal metric is

$$\|\dot{P}\|_{P, \text{AI}}^2 = \left\| R^{-1/2} \dot{R} R^{-1/2} \right\|_F^2 + 2 \text{tr}(K[R + R^{-1} - 2I]K^\top). \quad (3)$$

The first term measures visible geometry change. The second measures rotation of the visible subspace. Reference-valued modes have zero rotation cost and their admissible rotations form precisely the singular directions of the descriptor. Polarizing (3) gives a closed-form pair metric for arbitrary coupled perturbation directions. Its multi-direction matrix is a positive semidefinite Gram matrix. This metric cross-term measures the AIRM geometry of completed motion; it is distinct from the Hessian of the reduced minimum-distance value.

Contributions. The fixed-channel SPD theorem and the moving-channel pair metric are the two central results. The abstract, structured, and statistical theorems identify their mechanism and consequences.

1. *Exact SPD reduction and completion.* We prove that every full-column-rank SPD compression is a split-Hadamard submetry and derive the closed-form completion (2), its horizontal geometry, exact full-to-visible decision reduction, and closed-form prior–data shrinkage. A radius-conditioned minimax identity quantifies the remaining invisible ambiguity.
2. *Moving-channel geometry.* We identify the gauge quotient and all rank strata of completed SPD geometry, and derive the closed-form mismatch-weighted pair metric whose diagonal is (3). Its Gram factorization gives exact rank, metric-orthogonality and differential-kernel criteria, together with moving-reference covariance and the rank-loss boundary.
3. *General mechanism.* We characterize metric submetries by attained fiber distance and universal radial-visible reduction. Complete horizontal leaves on Hadamard manifolds yield coherent isometric sections, exact proximal commutation, and solution-wise gradient-flow lifting.
4. *Implementability and recovery.* We characterize diagonal and block structured expressivity by relative interiors of conic images, with strict separation and facial certificates for distinct failure modes. Deterministic and finite-sample bounds then separate visible AIRM error from Grassmann subspace error.

Scope. The theory starts after a positive visible target has been declared. Fisher, Gauss–Newton, gradient covariance, damping, and user-specified audit targets provide different constructions of such a target; the paper does not identify them with one another. The moving-reference formula is a kinematic identity, not an optimizer evolution law. Likewise, exact reduction determines a canonical full representative of a visible decision but does not prescribe which visible decision an optimizer should make.

Organization. Section 2 positions the work. Section 3 establishes the maximal metric theorem, and section 4 gives its smooth global form. Sections 5 and 6 develop the explicit SPD realization and its rank-stratified pair metric. Section 7 treats structured families, and section 8 develops perturbation and sampling results.

2 Related Work

SPD geometry and matrix completion. The affine-invariant geometry of positive matrices is classical (Bhatia, 2007; Pennec et al., 2006). Positive definite and covariance completion usually infer missing matrix entries from sparsity or graph constraints (Dempster, 1972; Grone et al., 1984). Here the known object is the compression $A^\top PA$, and the missing object is the entire positive-definite fiber. The selected completion is defined by AIRM distance to a reference and is characterized globally as a horizontal section of a metric submetry.

Best approximation under AIRM has been studied for geodesically convex, unitarily invariant subsets through logarithmic and spectral reductions (Bhatia & Jain, 2014), and for Riemannian geodesic submanifolds (Lim, 2004). Totally geodesic AIRM submanifolds and their nearest-point projections have also been characterized directly inside the SPD manifold (Tumpach & Larotonda, 2024). Our information sheet is a particular congruence-transformed block submanifold of that kind. The additional object here is the transverse affine-compression fiber: we identify its unique nearest point to every reference, prove that the compression is a metric submetry, and derive universal radial-visible reduction. Thus total geodesy supports the construction but does not by itself supply the fiber-completion or variational statements.

The distinction can be stated theorem by theorem. Existing AIRM approximation results supply general uniqueness or explicit projections for specified convex or totally geodesic target sets. In Theorem 5.2, the constrained set is instead the generally noncompact fiber $\{P : A^\top PA = R\}$. Contraction of the compression gives the lower bound, while a transverse block section attains it for every target. The new package is the equality-attaining unique fiber lift, its coherent horizontal foliation and submetry consequences, followed by the moving-channel pullback metric of Theorem 6.2. We use the classical approximation and totally geodesic

theories as inputs to this positioning, rather than claiming that AIRM nearest-point uniqueness by itself is new.

Quotient geometries for fixed-rank positive-semidefinite matrices have also been developed for matrix means and optimization (Bonnabel & Sepulchre, 2010; Vandereycken et al., 2013; Massart & Absil, 2020). Those manifolds parameterize a rank- m PSD matrix. Our stratum instead fixes the rank of the displacement $P - P_0$, which may be indefinite while P remains positive definite, and its line element is the pullback of the ambient AIRM at $P_0 + (P - P_0)$. The resulting mismatch weight $R + R^{-1} - 2I$ and its reference-valued rotation kernel are specific to this completion geometry. Viewed only as a set, each fixed-inertia component of this displacement stratum is an open part of the standard fixed-rank symmetric matrix manifold. The contribution is therefore not the existence of that manifold alone; it is the global completion-descriptor diffeomorphism and the ambient-AIRM pullback, whose mismatch weight and reference-valued rotation kernel do not arise from the usual Euclidean fixed-rank metric.

Riemannian submersions and submetries. The differential geometry of Riemannian submersions starts from the horizontal-vertical equations of O’Neill (O’Neill, 1966). Metric submetries provide the corresponding ball-surjectivity language; Berestovskii and Guijarro (Berestovskii & Guijarro, 2000) characterize Riemannian submersions through this metric property, and Guijarro and Walschap (Guijarro & Walschap, 2011) delineate when submetries recover smooth submersions. Our abstract theorem uses this classical metric structure and adds a variational characterization: exact fiber distance is equivalent to lossless reduction for every monotone radial visible objective. The split-Hadamard result imposes global integrable horizontal geometry to obtain coherent shortest sections, while the SPD compression theorem verifies this structure and resolves the section in closed form.

Information geometry and natural gradient. Natural gradient equips statistical model space with Fisher geometry (Amari, 1998). Structured Fisher and Gauss–Newton approximations motivate K-FAC, EK-FAC, and related methods (Martens & Grosse, 2015; George et al., 2018; Martens, 2020); limitations of identifying Fisher and Hessian geometry are discussed by Kunstner, Balles, and Hennig (Kunstner et al., 2019). Mirror descent also admits an information-geometric interpretation (Raskutti & Mukherjee, 2015). Our focus is the observation layer beneath a chosen training geometry: which cotangent information is visible, which ambient completions are compatible, and which completion is selected by a reference.

Sufficient statistics contract information-geometric tensors and preserve them under stronger sufficiency conditions (Ay et al., 2015). That work concerns statistical models and Fisher-type tensors. Here the declared object is a finite-dimensional positive cometric, the observation is the linear compression $A^\top P A$, and the main question is its reference-dependent AIRM completion. Statistical sufficiency motivates the visibility language, while the completion and moving-channel line element address a different geometric problem. The AIRM covariance factor is also the Fisher geometry of the multivariate normal family up to convention-dependent scaling (Lovric et al., 2000); this explains the Gaussian interpretation but does not determine the reference-relative completion of an arbitrary declared visible cometric.

Metric learning and relative-entropy projections. Information-theoretic metric learning uses LogDet or relative-entropy projections under supervised constraints (Davis et al., 2007). Our Gaussian KL consequence is related, but the main object is the full fiber of an information compression. The metric-submetry theorem explains when every radial visible decision, not only a particular divergence projection, reduces without loss. Classical information projections minimize divergence under linear or convex constraints (Csiszar, 1975; Csiszar & Matus, 2003). Our Gaussian KL corollary is a finite-dimensional instance with a closed-form fiber projection; the AIRM shortest-completion and stratified line-element claims are separate from the divergence-projection theory.

Adaptive and structured preconditioning. AdaGrad, Adam, Shampoo, and related methods impose diagonal, block, matrix, or tensor structure on positive preconditioners (Duchi et al., 2011; Kingma & Ba, 2015; Gupta et al., 2018; Morwani et al., 2024). Optimal diagonal preconditioning and matrix-scaling viewpoints study related coordinate restrictions (Qu et al., 2025). We formulate a different certificate: whether

a structured positive family can realize a declared visible target. The answer is a relative-interior condition in a linear conic image, which distinguishes strict representability, boundary failure, and closed-cone infeasibility.

Convex certificates. The structured certificates use finite-dimensional convex separation, relative interiors, and semidefinite images (Rockafellar, 1970; Vandenberghe & Boyd, 1996; Boyd & Vandenberghe, 2004). These tools enter only after the full-SPD information map has been resolved; they quantify what is lost when the completion is restricted to an implementable optimizer family.

Subspace perturbation and matrix concentration. When the visible subspace is estimated from a covariance or Gram matrix, its recovery requires an eigengap and a principal-angle perturbation bound (Yu et al., 2015). Covariance concentration then controls the ambient operator error (Tropp, 2015). We combine these ingredients with the exact stratified line element to obtain a completed-geometry error bound that separates visible SPD error from subspace rotation.

3 Metric Foundations of Exact Information Reduction

We first separate exact information reduction from smooth or matrix-specific structure. Let (X, d_X) and (Y, d_Y) be arbitrary metric spaces and let $q : X \rightarrow Y$ be surjective.

Definition 3.1 (Metric submetry). The map q is a metric submetry if for every $x \in X$ and $r \geq 0$,

$$q(\overline{B}_X(x, r)) = \overline{B}_Y(q(x), r). \quad (4)$$

A submetry maps every closed ball onto the ball of the same radius. It is therefore stronger than a nonexpansive surjection: it guarantees that every visible displacement has an ambient lift with no excess distance.

Theorem 3.2 (Maximal exact information reduction). *The following statements are equivalent.*

(A) q is a metric submetry.

(B) For every $x \in X$ and $y \in Y$, the fiber distance is attained and exact:

$$d_Y(q(x), y) = \min_{q(z)=y} d_X(x, z). \quad (5)$$

(C) For every nondecreasing $\rho : [0, \infty) \rightarrow \mathbb{R} \cup \{+\infty\}$, every $\phi : Y \rightarrow \mathbb{R} \cup \{+\infty\}$, and every $x \in X$,

$$\boxed{\inf_{z \in X} \{\rho(d_X(x, z)) + \phi(q(z))\} = \inf_{y \in Y} \{\rho(d_Y(q(x), y)) + \phi(y)\}.} \quad (6)$$

Thus metric submetries are exactly the observation maps for which every reference-radial, visible-functional decision problem reduces without loss.

The proof, given in section A, is elementary but sharp. Hard ball costs and singleton visible losses recover the full ball-surjectivity condition, so the theorem cannot be enlarged while preserving universal exact reduction.

The theorem determines optimal values but does not choose one lift when a fiber contains several nearest points. Canonical completion requires a coherent family of sections. Suppose q is 1-Lipschitz and for every $x \in X$ a section $s_x : Y \rightarrow X$ is given such that

$$q \circ s_x = \text{id}_Y, \quad s_x(q(x)) = x, \quad s_{s_x(y)} = s_x. \quad (7)$$

The last identity says that moving inside one selected information sheet does not change the sheet.

For a function F on a metric space, we use

$$e_\tau F(x) = \inf_z \left\{ F(z) + \frac{d(x, z)^2}{2\tau} \right\}$$

for its Moreau envelope and $\text{prox}_{\tau F}(x)$ for the corresponding minimizer when it is unique.

Theorem 3.3 (Coherent exact reduction). *Under (7), the following are equivalent:*

(A) *every s_x is an isometric embedding;*

(B) *for every $x \in X$ and $y \in Y$,*

$$d_Y(q(x), y) = d_X(x, s_x(y)) = \min_{q(z)=y} d_X(x, z); \quad (8)$$

(C) *the equality (6) holds and s_x maps every reduced minimizer to a full minimizer.*

Whenever these conditions hold, q is a metric submetry. If the nearest point in every fiber is unique, the splitting is rigid.

If Y is complete $CAT(0)$, ϕ is proper, lower semicontinuous, geodesically convex, and bounded below, and $\tau > 0$, then

$$\boxed{\text{prox}_{\tau(\phi \circ q)}^{X, \text{can}}(x) = s_x(\text{prox}_{\tau\phi}^Y(q(x)))}. \quad (9)$$

Coherent proximal iterations are therefore exact lifts of the visible iterations; under rigidity, the full proximal point is unique.

Remark 3.4 (Three distinct requirements). Surjectivity provides visible representatives. Submetry provides exact distance. Coherence provides a reusable canonical sheet. None of these properties follows from the others without the stated assumptions.

4 Split-Hadamard Information Maps

We now identify a smooth global condition that produces rigid coherent information sheets. Let (M, g) and (B, h) be finite-dimensional Hadamard manifolds, and let $q : M \rightarrow B$ be a surjective Riemannian submersion. Its vertical and horizontal distributions are

$$\mathcal{V}_x = \ker Dq(x), \quad \mathcal{H}_x = \mathcal{V}_x^\perp.$$

Definition 4.1 (Split-Hadamard information map).

The map q is split-Hadamard if $\mathcal{H}$ is integrable and every maximal horizontal leaf is complete and totally geodesic.

This condition rules out holonomy between visible and invisible directions. It does not require the fibers themselves to be linear or compact.

Theorem 4.2 (Global split-Hadamard information theorem). *Let $q : M \rightarrow B$ be split-Hadamard. For every $x \in M$, let L_x be the horizontal leaf through x . Then:*

(i) *The restriction $q|_{L_x} : L_x \rightarrow B$ is a global Riemannian isometry. Its inverse*

$$s_x : B \rightarrow L_x \subset M$$

is the unique horizontal isometric totally geodesic section satisfying $s_x(q(x)) = x$.

(ii) *The map q is a metric submetry and s_x is the unique shortest completion:*

$$s_x(b) = \arg \min_{q(z)=b} d_M(x, z), \quad d_M(x, s_x(b)) = d_B(q(x), b). \quad (10)$$

(iii) *Universal radial reduction (6) holds. Every reduced minimizer lifts through s_x ; if ρ is strictly increasing on its finite domain, these are all finite-valued full minimizers.*

(iv) Moreau envelopes and proximal maps commute exactly:

$$e_\tau^M(\phi \circ q)(x) = e_\tau^B \phi(q(x)), \quad (11)$$

$$\text{prox}_\tau^M(\phi \circ q)(x) = s_x(\text{prox}_\tau^B(q(x))) \quad (12)$$

for proper lower-semicontinuous geodesically convex bounded-below ϕ . The visible proximal point is unique by Hadamard convexity; the full proximal point is unique because equality requires both that visible point and its unique shortest lift.

(v) If $\phi \in C^1(B)$, then

$$\text{grad}_M(\phi \circ q)(z) = (Dq(z)|_{\mathcal{H}_z})^{-1} \text{grad}_B \phi(q(z)). \quad (13)$$

Consequently, every solution $y : I \rightarrow B$ of $\dot{y} = -\text{grad}_B \phi(y)$ with $y(0) = q(x)$ lifts on the same existence interval to the solution $z = s_x \circ y$ of $\dot{z} = -\text{grad}_M(\phi \circ q)(z)$ with $z(0) = x$. If $\text{grad}_B \phi$ is locally Lipschitz, the corresponding maximal solutions are unique.

The proof in section B has two ingredients. Completeness extends every horizontal lift of a visible geodesic, and Hadamard uniqueness makes the restriction to a horizontal leaf globally injective. Equality in Riemannian-submersion length contraction forces a shortest ambient geodesic to be horizontal, yielding uniqueness of the nearest completion.

Corollary 4.3 (Rigid coherent quotient). *The sections s_x satisfy (7); hence every split-Hadamard map is a rigid coherent metric quotient in the sense of Theorem 3.3.*

The abstract theorem is useful only after its hypotheses have been verified. The next section proves them globally for SPD information compression and, in addition, resolves the section in closed form.

5 SPD Information Compression and Canonical Completion

Let $A \in \mathbb{R}^{d \times m}$ have full column rank. The associated information compression is

$$\pi_A : \mathbb{S}_{++}^d \rightarrow \mathbb{S}_{++}^m, \quad \pi_A(P) = A^\top P A. \quad (14)$$

For $R, S \in \mathbb{S}_{++}^m$, write

$$R \#_t S = R^{1/2} (R^{-1/2} S R^{-1/2})^t R^{1/2}, \quad 0 \leq t \leq 1, \quad (15)$$

for AIRM geodesic interpolation. For a reference $P_0 \in \mathbb{S}_{++}^d$, write

$$R_0 = A^\top P_0 A, \quad U_0 = P_0^{1/2} A R_0^{-1/2}, \quad U_0^\top U_0 = I_m.$$

For $R \in \mathbb{S}_{++}^m$, define

$$\begin{aligned} \Psi_{P_0, A}(R) &= P_0^{1/2} \left[I + U_0 (R_0^{-1/2} R R_0^{-1/2} - I) U_0^\top \right] P_0^{1/2} \\ &= P_0 + P_0 A R_0^{-1} (R - R_0) R_0^{-1} A^\top P_0. \end{aligned} \quad (16)$$

This expression is invariant under the visible coordinate change $(A, R) \mapsto (AB, B^\top R B)$, $B \in GL(m)$.

Proposition 5.1 (A reference is necessary). *Let $W \subsetneq E$ and $R \in \mathbb{S}_{++}(W)$. There is no rule depending only on (W, R) that selects a full $P \in \mathbb{S}_{++}(E)$, restricts to R on W , and is invariant under every linear automorphism of E that fixes W pointwise. Canonical ambient completion therefore requires additional structure such as P_0 .*

Theorem 5.2 (Complete SPD information-compression theorem). *The map π_A and the section $\Psi_{P_0, A}$ satisfy the following.*

(A) Contraction and submetry. For all $P, Q \in \mathbb{S}_{++}^d$,

$$d_{\text{AI}}(A^\top P A, A^\top Q A) \leq d_{\text{AI}}(P, Q), \quad (17)$$

and for every $r \geq 0$,

$$\pi_A(\overline{B}_{\text{AI}}(P_0, r)) = \overline{B}_{\text{AI}}(R_0, r). \quad (18)$$

(B) Unique canonical completion. For every $R \in \mathbb{S}_{++}^m$,

$$\boxed{\Psi_{P_0,A}(R) = \arg \min_{A^\top P A = R} d_{\text{AI}}(P_0, P), \quad d_{\text{AI}}(P_0, \Psi_{P_0,A}(R)) = d_{\text{AI}}(R_0, R).} \quad (19)$$

The minimizer is unique.

(C) Horizontal geometry. The section $\Psi_{P_0,A}$ is an isometric totally geodesic embedding. The map π_A is a Riemannian submersion; at P , the inverse of $D\pi_A(P)|_{\mathcal{H}_P}$ is

$$h \mapsto D\Psi_{P,A}(A^\top P A)[h]. \quad (20)$$

Hence π_A is split-Hadamard and visible AIRM geodesics have unique horizontal ambient lifts.

(D) Invisible-fiber minimax geometry. If $m < d$, the fiber $\mathcal{F}_A(R) = \{P : A^\top P A = R\}$ has infinite AIRM diameter. With $P_\star = \Psi_{P_0,A}(R)$ and

$$\mathcal{F}_A(R; B) = \mathcal{F}_A(R) \cap \overline{B}_{\text{AI}}(P_\star, B),$$

one has

$$\inf_{Q \in \mathbb{S}_{++}^d} \sup_{P \in \mathcal{F}_A(R; B)} d_{\text{AI}}(Q, P) = B, \quad (21)$$

and $P_\star$ is the unique minimax center.

(E) Exact radial decision reduction. Let $\mathcal{A}$ be any family of admissible (P_0, A) , let $m(A)$ denote the number of columns of A , allowing different visible dimensions, let ρ be nondecreasing, and let φ be any extended-real visible functional. Then

$$\begin{aligned} & \inf_{\substack{(P_0, A) \in \mathcal{A} \\ P \in \mathbb{S}_{++}^d}} \{ \rho(d_{\text{AI}}(P_0, P)) + \varphi(P_0, A, A^\top P A) \} \\ &= \inf_{\substack{(P_0, A) \in \mathcal{A} \\ R \in \mathbb{S}_{++}^{m(A)}}} \{ \rho(d_{\text{AI}}(A^\top P_0 A, R)) + \varphi(P_0, A, R) \}. \end{aligned} \quad (22)$$

Every reduced minimizer has the canonical full lift $P = \Psi_{P_0,A}(R)$. If ρ is strictly increasing on its finite domain, every finite-valued full minimizer is canonical.

The closed form requires no constrained SPD optimization. A stable implementation forms $R_0 = A^\top P_0 A$, uses Cholesky or symmetric solves instead of explicit inverses, and applies the resulting rank- m update in (16). Dense assembly costs $O(d^2 m + d m^2 + m^3)$, while a low-rank representation avoids materializing the full update. Numerical sensitivity is governed by the conditioning of R_0 ; recovery of the descriptor subspace additionally becomes ill conditioned as an eigenvalue of the normalized target approaches the reference value one, as quantified in (42).

The set $\mathcal{F}_A(R; B)$ is a local AIRM ambiguity model centered at the reference-selected completion $P_\star$. Thus (21) is a radius-conditioned nonidentifiability statement: it does not define a reference-free uncertainty set or a reference-free minimax center.

Corollary 5.3 (Visible update quotient and invisible drift). *Let $\alpha = A c \neq 0$ be a visible covector and let $u_P = -P \alpha$. For every P satisfying $A^\top P A = R$,*

$$A^\top u_P = -R c. \quad (23)$$

Thus the class of the update modulo $\ker A^\top$ is fixed. Conversely, for every $n \in \ker A^\top$, there exists a feasible P such that

$$u_P = u_{\Psi_{P_0,A}(R)} + n. \quad (24)$$

The canonical completion is the unique AIRM-nearest representative and has no invisible component in the P_0 -orthogonal splitting.

Remark 5.4 (What is selected). The visible target R is an input to the theorem. The result selects the least AIRM deformation of P_0 compatible with that target. It does not assert that a first-order oracle uniquely determines R , nor that the reference P_0 is reference-free.

The same completion also has an information-theoretic characterization. For Gaussians it can equivalently be read from the KL chain rule as retaining the reference conditional law on invisible coordinates. We include it as a consequence and alternate characterization of the AIRM completion, rather than as an independent divergence-projection novelty claim.

Corollary 5.5 (Gaussian KL completion). *Viewing P as the covariance of a zero-mean Gaussian, the problem*

$$\min_{A^\top P A = R} D_{\text{KL}}(N(0, P) \parallel N(0, P_0)) \quad (25)$$

has the unique minimizer $\Psi_{P_0, A}(R)$.

For a fixed information channel, exact decision reduction yields an explicit regularized update. Let $R_0 = A^\top P_0 A$, let $\hat{R} \in \mathbb{S}_{++}^m$ be a positive data-visible target, and let $0 < \tau < 1$.

Corollary 5.6 (Prior-data geodesic completion). *The full-SPD objective*

$$(1 - \tau)d_{\text{AI}}(P_0, P)^2 + \tau d_{\text{AI}}(A^\top P A, \hat{R})^2 \quad (26)$$

has the unique minimizer

$$P_\tau = \Psi_{P_0, A}(R_0 \#_\tau \hat{R}). \quad (27)$$

If $D = d_{\text{AI}}(R_0, \hat{R})$, then

$$d_{\text{AI}}(P_0, P_\tau) = \tau D, \quad d_{\text{AI}}(A^\top P_\tau A, \hat{R}) = (1 - \tau)D,$$

and the minimum objective value is $\tau(1 - \tau)D^2$. Perturbing the visible target from R_1 to R_2 changes the completed solution by at most $\tau d_{\text{AI}}(R_1, R_2)$.

The proofs of Theorems 5.1 to 5.3, 5.5 and 5.6 are in section C. Their common mechanism is that orthogonal compression contracts the whitened Frobenius norm, while the section (16) attains equality.

6 Gauge, Rank Strata, and Moving Information

The previous section fixes A . We now let the information channel move. Relative to P_0 , the pair (A, R) has a change-of-visible-coordinates gauge

$$(A, R) \cdot B = (AB, B^\top RB), \quad B \in GL(m).$$

First define the total, generally redundant descriptor space

$$\tilde{\mathfrak{D}}_m(P_0) = \{(A, R) : \text{rank } A = m, R \in \mathbb{S}_{++}^m\} / GL(m), \quad (28)$$

and the everywhere-defined completion parameterization

$$\tilde{\Gamma}_{m, P_0}([A, R]) = \Psi_{P_0, A}(R). \quad (29)$$

When $m < d$, it is redundant if some visible modes equal their reference values, because those modes can rotate without changing the completion. For $m = d$, the Grassmann factor is a point and this rotational redundancy is absent. The regular subset is

$$\mathfrak{D}_m^{\text{reg}}(P_0) = \{(A, R) : \text{rank } A = m, R \in \mathbb{S}_{++}^m, \det(R - A^\top P_0 A) \neq 0\} / GL(m), \quad (30)$$

and the rank stratum

$$\mathfrak{S}_m(P_0) = \{P \in \mathbb{S}_{++}^d : \text{rank}(P - P_0) = m\}. \quad (31)$$

Theorem 6.1 (Completion bundle and complete rank stratification). *The map*

$$\Gamma_{m,P_0}([A, R]) = \Psi_{P_0,A}(R) \quad (32)$$

is a diffeomorphism from $\mathfrak{D}_m^{\text{reg}}(P_0)$ onto $\mathfrak{S}_m(P_0)$. Its inverse descriptor is intrinsic:

$$\text{range}(A) = P_0^{-1} \text{range}(P - P_0), \quad R = A^\top P A, \quad (33)$$

up to the stated $GL(m)$ action. The strata exhaust and close according to

$$\mathbb{S}_{++}^d = \bigsqcup_{m=0}^d \mathfrak{S}_m(P_0), \quad \overline{\mathfrak{S}_m(P_0)}^{\mathbb{S}_{++}^d} = \bigcup_{r=0}^m \mathfrak{S}_r(P_0). \quad (34)$$

The pullback AIRM seminorm on the total descriptor space resolves the cost of changing the visible target and the visible subspace. It becomes a genuine quotient metric on the regular subset, and on the entire descriptor space when $m = d$. Normalize a representative by $A^\top P_0 A = I_m$, use the horizontal gauge $A^\top P_0 \dot{A} = 0$, choose N so that $[P_0^{1/2} A, N]$ is orthogonal, and put $K = N^\top P_0^{1/2} \dot{A}$.

Theorem 6.2 (Closed-form completion-pair metric and regular quotient geometry). *At any normalized representative of $\tilde{\mathfrak{D}}_m(P_0)$, represent two horizontal quotient directions by*

$$\xi_i = (H_i, K_i), \quad H_i \in \mathbb{S}^m, \quad K_i = N^\top P_0^{1/2} \dot{A}_i, \quad i \in \{1, 2\}.$$

Then the pullback AIRM pair kernel is

$$\boxed{\mathfrak{G}_R^{\text{comp}}(\xi_1, \xi_2) := \left\langle D\tilde{\Gamma}_{m,P_0}[\xi_1], D\tilde{\Gamma}_{m,P_0}[\xi_2] \right\rangle_{\tilde{\Gamma}_{m,P_0}, \text{AI}} = \text{tr}(R^{-1} H_1 R^{-1} H_2) + 2 \text{tr}(K_1 \Xi(R) K_2^\top),} \quad (35)$$

where

$$\Xi(R) = (R - I)R^{-1}(R - I) = R + R^{-1} - 2I \succeq 0. \quad (36)$$

In particular, for $\xi = (\dot{R}, K)$,

$$\boxed{\left\| D\tilde{\Gamma}_{m,P_0}[\dot{A}, \dot{R}] \right\|_{\tilde{\Gamma}_{m,P_0}, \text{AI}}^2 = \mathfrak{G}_R^{\text{comp}}(\xi, \xi) = \left\| R^{-1/2} \dot{R} R^{-1/2} \right\|_F^2 + 2 \text{tr}(K \Xi(R) K^\top),} \quad (37)$$

so visible-target and subspace-motion blocks are orthogonal.

More explicitly, define the completion feature

$$\Phi_R(\xi) = \left(R^{-1/2} H R^{-1/2}, \sqrt{2} K (R - I) R^{-1/2} \right). \quad (38)$$

Then $\mathfrak{G}_R^{\text{comp}}(\xi_i, \xi_j) = \langle \Phi_R(\xi_i), \Phi_R(\xi_j) \rangle_F$ in the direct-sum Frobenius space. Hence for any directions $\xi_1, \dots, \xi_k$, the pair matrix

$$\mathbf{G} = [\mathfrak{G}_R^{\text{comp}}(\xi_i, \xi_j)]_{i,j=1}^k \succeq 0, \quad \text{rank } \mathbf{G} = \dim \text{span}\{\Phi_R(\xi_i) : 1 \leq i \leq k\}. \quad (39)$$

Moreover,

$$\mathfrak{G}_R^{\text{comp}}(\xi_i, \xi_j) = 0 \iff \Phi_R(\xi_i) \perp \Phi_R(\xi_j). \quad (40)$$

The exact horizontal differential kernel of the total parameterization is

$$\ker D\tilde{\Gamma}_{m,P_0} = \{(\dot{A}, 0) : K(R - I) = 0\}. \quad (41)$$

If $m < d$, the pullback is nondegenerate precisely on the regular subset $\det(R - I) \neq 0$, where $\tilde{\Gamma}_{m,P_0} = \Gamma_{m,P_0}$ is the diffeomorphism of Theorem 6.1. At singular descriptors, rotation of a reference-valued visible mode is

unidentifiable from the completed matrix. If $m = d$, there is no subspace-rotation block: the visible AIRM term is nondegenerate for every R , and the total completion map is a diffeomorphism onto $\mathbb{S}_{++}^d$. Thus descriptor singularity occurs only when $m < d$; the ambient SPD manifold and its AIRM remain smooth and nondegenerate in every case. If $\text{dist}(\text{spec}(R), 1) \geq \zeta > 0$ and $0 < a \leq \lambda(R) \leq b$, then subspace motion is quantitatively controlled:

$$\begin{aligned} \left\| R^{-1/2} \dot{R} R^{-1/2} \right\|_F^2 + \frac{2\zeta^2}{b} \|K\|_F^2 &\leq \|\dot{P}\|_{P, \text{AI}}^2 \\ &\leq \left\| R^{-1/2} \dot{R} R^{-1/2} \right\|_F^2 + 2 \max_{x \in [a, b]} \frac{(x-1)^2}{x} \|K\|_F^2. \end{aligned} \quad (42)$$

Corollary 6.3 (Reduced completion value and completed motion). *At fixed P_0 , in the normalized chart $A^\top P_0 A = I$, the reduced minimum-distance value is*

$$\bar{E}(A, R) := \min_{A^\top P A = R} \frac{1}{2} d_{\text{AI}}(P_0, P)^2 = \frac{1}{2} d_{\text{AI}}(I, R)^2. \quad (43)$$

It contains no independent information-subspace coordinate. Consequently, holding R fixed, all first and mixed parameter derivatives generated by horizontal variations of A vanish. Exact completion has already removed the need for an implicit vertical stationarity solve: the minimizing ambient matrix is the explicit section $P = \Psi_{P_0, A}(R)$. The pullback pair metric (35) is a different second-order object: it measures motion of this minimizing matrix rather than curvature of the reduced value.

Remark 6.4 (Reduced value versus completion motion). The distinction is already visible for $d = 2$, $m = 1$, $P_0 = I$, and a scalar target $r > 0$, $r \neq 1$. With $u(\theta) = (\cos \theta, \sin \theta)^\top$, the canonical completion is

$$P_\star(\theta) = I + (r - 1)u(\theta)u(\theta)^\top.$$

Its reduced minimum-distance value is the constant $\frac{1}{2} \log^2 r$, whereas

$$\|\dot{P}_\star(0)\|_{P_\star(0), \text{AI}}^2 = \frac{2(r-1)^2}{r} > 0.$$

Thus vanishing subspace derivatives of the optimized value coexist with nonzero metric motion of its canonical minimizer. A frozen-relaxed second-variation decomposition may likewise contain nonzero direct and Schur terms that cancel; their split depends on the chosen trivialization of the moving constraint. The completion kernel itself is analytic for every $R \in \mathbb{S}_{++}^m$, including repeated eigenvalues, and its degeneracy is exactly the descriptor non-identifiability in (41).

The same identity has a gauge-covariant moving-reference form. Let $P_{0,t} = G_t G_t^\top$, define

$$\begin{aligned} R_{0,t} &= A_t^\top P_{0,t} A_t, \quad U_t = G_t^\top A_t R_{0,t}^{-1/2}, \quad \bar{R}_t = R_{0,t}^{-1/2} R_t R_{0,t}^{-1/2}, \\ Q_t &= I + U_t(\bar{R}_t - I)U_t^\top, \quad \Omega_t = G_t^{-1} \dot{G}_t, \quad \mathfrak{D}_t Q_t = \dot{Q}_t + \Omega_t Q_t + Q_t \Omega_t^\top. \end{aligned}$$

Proposition 6.5 (Covariant moving-reference identity).

For $P_t = \Psi_{P_0, A_t}(R_t) = G_t Q_t G_t^\top$,

$$\boxed{\|\dot{P}_t\|_{P_t, \text{AI}}^2 = \left\| Q_t^{-1/2} (\mathfrak{D}_t Q_t) Q_t^{-1/2} \right\|_F^2.} \quad (44)$$

Under $G'_t = G_t O_t$ with O_t orthogonal,

$$Q'_t = O_t^\top Q_t O_t, \quad \mathfrak{D}'_t Q'_t = O_t^\top (\mathfrak{D}_t Q_t) O_t.$$

Hence (44) is independent of the moving frame.

For a fixed reference and a fixed-rank curve, integrating (37) gives the exact energy decomposition

$$\begin{aligned} \int_0^T \|\dot{P}_t\|_{P_t, \text{AI}}^2 dt &= \int_0^T \left\| R_t^{-1/2} \dot{R}_t R_t^{-1/2} \right\|_F^2 dt \\ &\quad + 2 \int_0^T \text{tr}(K_t \Xi(R_t) K_t^\top) dt. \end{aligned} \quad (45)$$

Corollary 6.6 (Boundary of continuous rank loss). *Suppose $P_t = P_0^{1/2}[I + U_t(R_t - I)U_t^\top]P_0^{1/2}$ has active rank m for $t < t_0$ and converges to a point of rank at most $m - r$ relative to P_0 . Then at least r singular values of $R_t - I$ converge to zero. Thus continuous rank loss can occur only through visible modes approaching the reference geometry.*

The proofs are collected in section D. The line element is a kinematic identity on the completion bundle. A learning or control law would require an additional evolution rule for (P_0, A, R) .

7 Structured Expressivity Certificates

The full-SPD channel can realize every $R \in \mathbb{S}_{++}^m$. Implementable optimizers often restrict P to a coordinate, block, Kronecker, or other structured family. For a family $\mathcal{F} \subset \mathbb{S}_{++}^d$ equipped with a within-family distance $d_{\mathcal{F}}$, define the visible restricted complexity

$$D_{\mathcal{F}}^{\text{vis}}(A, R; P_0) = \inf_{\substack{P \in \mathcal{F} \\ A^\top P A = R}} d_{\mathcal{F}}(P_0, P), \quad (46)$$

with value $+\infty$ when no structured completion exists. This section characterizes finiteness for diagonal and block families.

7.1 Diagonal geometry

Let $a_j^\top$ be the j -th row of $A \in \mathbb{R}^{d \times m}$ and set $G_j = a_j a_j^\top \in \mathbb{S}_+^m$. A positive diagonal cometric $D(p) = \text{diag}(p_1, \dots, p_d)$ has visible image

$$A^\top D(p) A = \sum_{j=1}^d p_j G_j. \quad (47)$$

Let $\mathcal{K}_{\text{diag}} = \text{cone}(G_1, \dots, G_d)$.

Theorem 7.1 (Diagonal expressivity and facial alternative).

Let $D_0 = \text{diag}(p_1^0, \dots, p_d^0)$ with $p_j^0 > 0$, and define

$$D_{\text{diag}}^{\text{vis}} = \inf_{\substack{p_j > 0 \\ \sum_j p_j G_j = R}} \left(\sum_{j=1}^d \log^2 \frac{p_j}{p_j^0} \right)^{1/2}. \quad (48)$$

Then $D_{\text{diag}}^{\text{vis}} < \infty$ if and only if

$$R \in \text{ri } \mathcal{K}_{\text{diag}}, \quad (49)$$

equivalently, $R = \sum_j p_j G_j$ for some $p_j > 0$. When finite, the infimum is attained. If the G_j are linearly independent, the feasible diagonal cometric is unique.

There are two distinct failure certificates.

(i) If $R \notin \mathcal{K}_{\text{diag}}$, there exists symmetric Y such that

$$\text{tr}(YR) < 0, \quad \text{tr}(YG_j) \geq 0 \quad \forall j.$$

(ii) If $R \in \mathcal{K}_{\text{diag}} \setminus \text{ri} \mathcal{K}_{\text{diag}}$, there exists symmetric Y such that

$$\text{tr}(YR) = 0, \quad \text{tr}(YG_j) \geq 0 \quad \forall j, \quad \text{tr}(YG_{j_*}) > 0 \quad \text{for some } j_*.$$

The first strictly separates an infeasible target; the second exposes the proper face containing a boundary target that requires some diagonal weights to vanish.

Example 7.2 (An end-to-end visible training channel). Let the rows of $A \in \mathbb{R}^{3 \times 2}$ be

$$a_1^\top = (1, 0), \quad a_2^\top = (0, 1), \quad a_3^\top = (1, 1).$$

Take $P_0 = I_3$ and the visible target $R = I_2$. Since

$$R_0 = A^\top A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

the full-SPD canonical completion is

$$P_\star = \Psi_{I_3, A}(I_2) = \frac{1}{9} \begin{pmatrix} 8 & -1 & -2 \\ -1 & 8 & -2 \\ -2 & -2 & 5 \end{pmatrix}, \quad A^\top P_\star A = I_2.$$

Thus the visible request has a unique nearest full-SPD realization. Within the diagonal family, however, it requires the semidefinite coefficient vector $p = (1, 1, 0)$; no strictly positive diagonal geometry realizes it. The matrix

$$Y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is a facial certificate: $\text{tr}(YR) = 0$, $\text{tr}(YG_1) = \text{tr}(YG_2) = 0$, and $\text{tr}(YG_3) = 2$, so it is a nontrivial functional on $\text{span} \mathcal{K}_{\text{diag}}$.

For the visible covector $\alpha = Ae_1 = (1, 0, 1)^\top$, the canonical update is

$$u_\star = -P_\star \alpha = \left(-\frac{2}{3}, \frac{1}{3}, -\frac{1}{3} \right)^\top, \quad A^\top u_\star = -e_1.$$

Every feasible full geometry has this same visible update, while its invisible component may differ by any vector in $\ker A^\top = \text{span}\{(1, 1, -1)^\top\}$. The example therefore displays, in one calculation, exact visible reduction, canonical full completion, invisible drift, and a structured-family boundary certificate.

7.2 Block geometry

Partition the rows of A into blocks A_b . A block cometric has the form $P = \text{blockdiag}(P_1, \dots, P_B)$, $P_b \succ 0$, and visible image

$$A^\top P A = \sum_b A_b^\top P_b A_b. \quad (50)$$

Define

$$\mathcal{K}_{\text{block}} = \left\{ \sum_b A_b^\top P_b A_b : P_b \succeq 0 \right\}.$$

Theorem 7.3 (Block expressivity).

For a block reference $P_0 = \text{blockdiag}(P_1^0, \dots, P_B^0)$, the quantity

$$D_{\text{block}}^{\text{vis}} = \inf_{\substack{P_b \succ 0 \\ \sum_b A_b^\top P_b A_b = R}} \left(\sum_b d_{\text{AI}}(P_b^0, P_b)^2 \right)^{1/2} \quad (51)$$

is finite if and only if $R \in \text{ri } \mathcal{K}_{\text{block}}$, equivalently if positive definite blocks realize R . When finite, the infimum is attained.

The cone $\mathcal{K}_{\text{block}}$ is closed. If $R \notin \mathcal{K}_{\text{block}}$, there is symmetric Y such that

$$\text{tr}(YR) < 0, \quad A_b Y A_b^\top \succeq 0 \quad \forall b.$$

If $R \in \mathcal{K}_{\text{block}} \setminus \text{ri } \mathcal{K}_{\text{block}}$, there is a facial certificate with

$$\text{tr}(YR) = 0, \quad A_b Y A_b^\top \succeq 0 \quad \forall b, \quad A_{b_*} Y A_{b_*}^\top \neq 0 \quad \text{for some } b_*.$$

The proofs in section E use the identity that a linear map sends the relative interior of a finite-dimensional convex cone onto the relative interior of its image. The distinction between strict separation and facial exposure is essential: targets on a proper face are realizable by semidefinite structured geometries while remaining unreachable by strictly positive optimizer blocks. In both facial alternatives, nontriviality is measured on the linear span of the image cone; ambient nonzero Y alone is insufficient because it may annihilate the entire cone.

The certificates have direct finite-dimensional feasibility forms. For the diagonal boundary case, scaling permits the normalization $\sum_j \text{tr}(Y G_j) = 1$, leaving linear equalities and inequalities in Y ; strict infeasibility can be normalized by $\text{tr}(YR) \leq -1$. For block geometry, the corresponding constraints $A_b Y A_b^\top \succeq 0$ together with $\sum_b \text{tr}(A_b Y A_b^\top) = 1$ form an SDP feasibility problem. These formulations make the certificates computable in principle; algorithmic complexity is outside the present characterization. In finite precision, the normalization equality, cone residual, and strict separation margin should be reported together: a small residual with a vanishing margin signals proximity to a proper face, whereas the exact alternatives in the theorems concern exact feasibility.

Kronecker, low-rank, and tensor families require separate image geometry and are not covered by the diagonal/block theorems.

8 Recovery of Visible and Completed Geometry

The isometric fixed-channel section transfers visible SPD error exactly. When the information subspace is estimated, the stratified line element adds a second, mismatch-weighted Grassmann term.

All results below are stated after congruence whitening the reference to $P_0 = I$; AIRM congruence invariance transports them to an arbitrary fixed reference. In these coordinates, for an orthonormal frame $U \in \text{St}(d, m)$, write

$$\mathfrak{C}(U, R) = I + U(R - I)U^\top. \quad (52)$$

Two observation regimes must be distinguished. The exact visible space in section 1 is the span of the observed covectors. The known-subspace result below conditions on any such fixed space. The unknown-subspace result instead adopts a population second-moment model and defines its target space as the leading m -dimensional eigenspace of Σ . This spectral target agrees with the population covector support under a rank- m population model; a finite observed span equals it only when the sampled covectors span that support. With nonzero spectral tails it instead represents a chosen principal approximation. No automatic identification of the two regimes is assumed.

For $W, \widetilde{W} \in \text{Gr}(m, d)$, let $d_{\text{Gr}}(W, \widetilde{W})$ be the Euclidean norm of their principal-angle vector. Define

$$\chi(R) = \|\Xi(R)\|_{\text{op}}^{1/2} = \max_{\lambda \in \text{spec}(R)} \frac{|\lambda - 1|}{\sqrt{\lambda}}. \quad (53)$$

Theorem 8.1 (Joint deterministic perturbation). *Let U, V be endpoint frames connected by a horizontal minimizing Grassmann geodesic, and express $R, S \in \mathbb{S}_{++}^m$ in the parallel endpoint frames. With reference I ,*

$$d_{\text{AI}}(\mathfrak{C}(U, R), \mathfrak{C}(V, S)) \leq d_{\text{AI}}(R, S) + \sqrt{2} \min\{\chi(R), \chi(S)\} d_{\text{Gr}}(W, \widetilde{W}). \quad (54)$$

The first term is visible geometry error. The second is subspace error, and it vanishes for reference-valued visible geometry.

The endpoint-frame condition is a gauge choice, not an additional statistical assumption. Given arbitrary orthonormal endpoint frames, align them by the orthogonal Procrustes factor associated with the minimizing Grassmann geodesic and conjugate the endpoint visible matrix accordingly. The statement then applies in the aligned coordinates, and its value is independent of the initial endpoint gauges.

For a known fixed visible subspace, a direct concentration theorem retains the intrinsic dimension. In an orthonormal coordinate frame of W , let $X_1, \dots, X_n \in \mathbb{R}^m$ be i.i.d. copies of X , define the uncentered second moment $R = \mathbb{E}[XX^\top] \succ 0$, assume $\|R^{-1/2}X\|_2 \leq L$ almost surely, and set $\hat{R} = n^{-1} \sum_i X_i X_i^\top$.

Theorem 8.2 (Known-subspace finite-sample recovery). *For $0 < \varepsilon \leq 1$, if*

$$n \geq \frac{8L^4}{\varepsilon^2} \left(m \log 9 + \log \frac{2}{\delta} \right), \quad (55)$$

then with probability at least $1 - \delta$,

$$e^{-\varepsilon} R \preceq \hat{R} \preceq e^\varepsilon R, \quad d_{\text{AI}}(\mathfrak{C}(U, R), \mathfrak{C}(U, \hat{R})) \leq \sqrt{m} \varepsilon. \quad (56)$$

This self-contained net-Hoeffding bound is deliberately conservative. In fact

$$m = \mathbb{E} \left\| R^{-1/2} X \right\|_2^2 \leq L^2,$$

so the displayed sufficient sample size can have order m^3/ε^2 when only this envelope is retained. Matrix Chernoff or Bernstein inequalities can replace the net argument and often improve the dependence to a logarithmic dimension or an effective-rank factor under their corresponding variance or tail assumptions (Tropp, 2015). We use the elementary bound for transparency and do not claim an optimal statistical rate.

We next allow the visible eigenspace to be unknown. Let $1 \leq m < d$ and $\Sigma \in \mathbb{S}_+^d$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$, assume $\gamma = \lambda_m - \lambda_{m+1} > 0$, and let U span its leading m -dimensional eigenspace. Put

$$R = U^\top \Sigma U, \quad P = \mathfrak{C}(U, R).$$

For an estimator $\hat{\Sigma}$, let V span its leading eigenspace, $\hat{R} = V^\top \hat{\Sigma} V$, and $\hat{P} = \mathfrak{C}(V, \hat{R})$.

Theorem 8.3 (Unknown-subspace completed-geometry recovery). *Let*

$$\left\| \hat{\Sigma} - \Sigma \right\|_{\text{op}} \leq \delta < \min\{\gamma/2, \lambda_m\}, \quad s_\delta = \frac{\delta}{\gamma - \delta},$$

$$\varepsilon_{\text{vis}} = \delta + 2\sqrt{2}\lambda_1 s_\delta, \quad \eta_\delta = \frac{\varepsilon_{\text{vis}}}{\lambda_m} < 1,$$

and

$$\omega_\delta = \max_{x \in [\lambda_m - \delta, \lambda_1 + \delta]} \frac{|x - 1|}{\sqrt{x}}.$$

Then

$$\boxed{d_{\text{AI}}(P, \hat{P}) \leq \sqrt{m} \left[-\log(1 - \eta_\delta) + \sqrt{2}\omega_\delta \arcsin(s_\delta) \right]}. \quad (57)$$

The first term controls the aligned visible second moment; the second is the mismatch-weighted rotation of the estimated eigenspace.

Corollary 8.4 (A self-contained sample bound). *Let $X_1, \dots, X_n \in \mathbb{R}^d$ be i.i.d. copies of X , with uncentered second moment $\Sigma = \mathbb{E}[XX^\top]$, $\|X\|_2 \leq L$ almost surely, and $\hat{\Sigma} = n^{-1} \sum_i X_i X_i^\top$. Define*

$$\delta_n(\alpha) = L^2 \sqrt{\frac{2(d \log 9 + \log(2/\alpha))}{n}}. \quad (58)$$

With probability at least $1 - \alpha$, $\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \leq \delta_n(\alpha)$. If $\delta_n(\alpha)$ satisfies the hypotheses of Theorem 8.3, then (57) holds with $\delta = \delta_n(\alpha)$.

In particular, put

$$c_\gamma = 1 + \frac{8\sqrt{2}\lambda_1}{3\gamma}, \quad \delta_\star = \min \left\{ \frac{\gamma}{4}, \frac{\lambda_m}{2c_\gamma} \right\}. \quad (59)$$

The directly checkable sufficient condition

$$n \geq \frac{2L^4}{\delta_\star^2} \left(d \log 9 + \log \frac{2}{\alpha} \right) \quad (60)$$

implies $\delta_n(\alpha) \leq \delta_\star$, hence $s_\delta \leq 4\delta/(3\gamma)$ and $\eta_\delta \leq 1/2$. It therefore guarantees all hypotheses of Theorem 8.3.

The ambient dimension in (58) is the price of estimating the subspace with a self-contained bounded-i.i.d. net argument. Matrix concentration or effective-rank assumptions may improve this factor, but an eigengap or another identifiable subspace condition remains necessary. The reference I here is the whitened reference geometry, and Σ is an uncentered second moment; it coincides with covariance only when the mean is zero. Proofs are in section F.

9 Discussion

What the theory establishes. Exact information reduction has a precise hierarchy. At the metric level, submetry is necessary and sufficient for universal radial-visible reduction. Coherent sections select reusable information sheets. Split-Hadamard geometry turns those sheets into global horizontal submanifolds. SPD compression realizes the entire hierarchy in closed form and adds a rank quotient, a closed-form completion-pair metric, and computable structured certificates in finite-dimensional LP/SDP feasibility form.

Visibility and reference. A visible target does not determine an ambient geometry. The invisible fiber has infinite diameter, and even its radius-bounded minimax problem has risk equal to the radius. The canonical completion is therefore meaningful only relative to a declared P_0 . Its role is precise: preserve the reference on directions not forced to change by the visible target while realizing the visible target with minimum AIRM deformation.

Moving information. The mismatch weight $\Xi(R) = R + R^{-1} - 2I$ explains when subspace motion is observable in completed geometry. Rotating a mode whose visible geometry equals the reference changes no completed matrix. As a mode separates from the reference, the same rotation acquires positive cost. Continuous rank changes can therefore occur only through reference-valued modes. These are kinematic statements; choosing a trajectory requires an additional modeling, optimization, or control principle. The bilinear kernel also separates two second-order objects. The reduced minimum-distance value is independent of the normalized subspace coordinate at fixed R , while the minimizing completed matrix generally moves with that coordinate. Its motion is measured by the positive-semidefinite Gram form (35). Reduced-value curvature and pullback metric geometry therefore answer different questions.

Structured optimizers. Diagonal and block geometry are restrictions of the same visible map $P \mapsto A^\top P A$. Their relative-interior tests distinguish three scientifically different outcomes: strict representability, boundary representability requiring degeneracy, and infeasibility even after closure. A single residual value cannot make these distinctions; the conic certificates can.

Statistical interpretation. Known-subspace recovery is intrinsically m -dimensional. Estimating the subspace introduces an ambient second-moment problem and an eigengap. The exact line element then converts principal-angle error into completed AIRM error with a geometry-dependent weight. The resulting bound separates statistical uncertainty in R from uncertainty in the information channel itself. The self-contained rates use bounded i.i.d. observations and net-Hoeffding arguments; they establish consistency in the stated regime and are not rate-optimal claims.

Boundaries. The abstract theorem characterizes exactness but does not make an arbitrary observation map a submetry. Nonlinear, infinite-dimensional, multi-view, or non-AIRM models require separate verification. The SPD pair metric is specific to full-column-rank linear compression and is analytic across repeated eigenvalues. The structured certificate theorems cover diagonal and block families; Kronecker, low-rank, and tensor families have different image geometry. Finally, the paper assumes a positive visible target. Indefinite curvature must be damped, projected, or otherwise converted before this theory applies.

Downstream use. The paper supplies a geometric foundation for later questions about metric evolution, constrained metric paths, and intervention-based optimizer audits. Those questions may use the visible state (A, R) , the canonical completion $\Psi_{P_0, A}(R)$, and the completion-pair metric derived here. Their dynamical or causal assumptions are intentionally outside the present paper.

10 Reproducibility and Ethics Statements

Reproducibility. This is a theory paper. Every claim is stated as a definition, theorem, corollary, or explicitly bounded interpretation. Complete proofs are given in the appendices. The finite-sample results specify the boundedness, eigengap, confidence, and dimension dependence used by each conclusion. No empirical optimizer-performance claim is made.

Ethics and broader impact. The paper introduces no dataset, human-subject experiment, or deployed decision system. Its main interpretive risk is to treat a visible geometric certificate as a complete explanation of an optimizer. The theory identifies what a declared information channel constrains; memory, noise, higher-order oracles, discretization, and other mechanisms can produce additional motion.

References

- Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998. doi: 10.1162/089976698300017746.
- Nihat Ay, Jurgen Jost, Hong Van Le, and Lorenz Schwachhofer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162(1–2):327–364, 2015. doi: 10.1007/s00440-014-0574-8. URL <https://doi.org/10.1007/s00440-014-0574-8>.
- V. N. Berestovskii and Luis Guijarro. A metric characterization of riemannian submersions. *Annals of Global Analysis and Geometry*, 18(6):577–588, 2000. doi: 10.1023/A:1006683922481. URL <https://doi.org/10.1023/A:1006683922481>.
- Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.
- Rajendra Bhatia and Tanvi Jain. Approximation problems in the riemannian metric on positive definite matrices. *Annals of Functional Analysis*, 5(2):118–126, 2014. doi: 10.15352/afa/1396833507. URL <https://doi.org/10.15352/afa/1396833507>.
- Silvere Bonnabel and Rodolphe Sepulchre. Riemannian metric and geometric mean for positive semidefinite matrices of fixed rank. *SIAM Journal on Matrix Analysis and Applications*, 31(3):1055–1070, 2010. doi: 10.1137/080731347. URL <https://doi.org/10.1137/080731347>.
- Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004. URL <https://web.stanford.edu/~boyd/cvxbook/>.
- Imre Csiszar. I -Divergence Geometry of Probability Distributions and Minimization Problems. *The Annals of Probability*, 3(1):146–158, 1975. doi: 10.1214/aop/1176996454. URL <https://doi.org/10.1214/aop/1176996454>.
- Imre Csiszar and Frantisek Matus. Information projections revisited. *IEEE Transactions on Information Theory*, 49(6):1474–1490, 2003. doi: 10.1109/TIT.2003.810633. URL <https://doi.org/10.1109/TIT.2003.810633>.

-
- Jason V. Davis, Brian Kulis, Prateek Jain, Suvrit Sra, and Inderjit S. Dhillon. Information-theoretic metric learning. In *Proceedings of the 24th International Conference on Machine Learning*, pp. 209–216. ACM, 2007. doi: 10.1145/1273496.1273523. URL <https://doi.org/10.1145/1273496.1273523>.
- Arthur P. Dempster. Covariance selection. *Biometrics*, 28(1):157–175, 1972. doi: 10.2307/2528966.
- John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12:2121–2159, 2011. URL <https://jmlr.org/papers/v12/duchi11a.html>.
- Thomas George, César Laurent, Xavier Bouthillier, Nicolas Ballas, and Pascal Vincent. Fast approximate natural gradient descent in a Kronecker-factored eigenbasis. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL <https://arxiv.org/abs/1806.03884>.
- Robert Grone, Charles R. Johnson, Eduardo M. Sá, and Henry Wolkowicz. Positive definite completions of partial hermitian matrices. *Linear Algebra and its Applications*, 58:109–124, 1984. doi: 10.1016/0024-3795(84)90207-6.
- Luis Guijarro and Gerard Walschap. Submetries vs. submersions. *Revista Matemática Iberoamericana*, 27(2): 605–619, 2011. doi: 10.4171/RMI/648. URL <https://doi.org/10.4171/RMI/648>.
- Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned stochastic tensor optimization. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 1842–1850. PMLR, 2018. URL <https://proceedings.mlr.press/v80/gupta18a.html>.
- Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL <https://arxiv.org/abs/1412.6980>. arXiv:1412.6980.
- Frederik Kunstner, Lukas Balles, and Philipp Hennig. Limitations of the empirical fisher approximation for natural gradient descent. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/46a558d97954d0692411c861cf78ef79-Abstract.html>.
- Yongdo Lim. Best approximation in riemannian geodesic submanifolds of positive definite matrices. *Canadian Journal of Mathematics*, 56(4):776–793, 2004. doi: 10.4153/CJM-2004-035-5.
- Miroslav Lovric, Maung Min-Oo, and Ernst A. Ruh. Multivariate normal distributions parametrized as a riemannian symmetric space. *Journal of Multivariate Analysis*, 74(1):36–48, 2000. doi: 10.1006/jmva.1999.1853. URL <https://doi.org/10.1006/jmva.1999.1853>.
- James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020. URL <https://jmlr.org/papers/v21/martens15.html>.
- James Martens and Roger Grosse. Optimizing neural networks with Kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 2408–2417. PMLR, 2015. URL <https://proceedings.mlr.press/v37/martens15.html>.
- Estelle Massart and P.-A. Absil. Quotient geometry with simple geodesics for the manifold of fixed-rank positive-semidefinite matrices. *SIAM Journal on Matrix Analysis and Applications*, 41(1):171–198, 2020. doi: 10.1137/18M1231389. URL <https://doi.org/10.1137/18M1231389>.
- Depen Morwani, Itai Shapira, Nikhil Vyas, Eran Malach, Sham M. Kakade, and Lucas Janson. A new perspective on Shampoo’s preconditioner, 2024. URL <https://arxiv.org/abs/2406.17748>.
- Barrett O’Neill. The fundamental equations of a submersion. *Michigan Mathematical Journal*, 13(4):459–469, 1966. doi: 10.1307/mmj/1028999604. URL <https://doi.org/10.1307/mmj/1028999604>.
- Xavier Pennec, Pierre Fillard, and Nicholas Ayache. A riemannian framework for tensor computing. *International Journal of Computer Vision*, 66(1):41–66, 2006. doi: 10.1007/s11263-005-3222-z.

-
- Zhaonan Qu, Wenzhi Gao, Oliver Hinder, Yinyu Ye, and Zhengyuan Zhou. Optimal diagonal preconditioning. *Operations Research*, 73(3):1479–1495, 2025. doi: 10.1287/opre.2022.0592. URL <https://doi.org/10.1287/opre.2022.0592>.
- Garvesh Raskutti and Sayan Mukherjee. The information geometry of mirror descent. *IEEE Transactions on Information Theory*, 61(3):1451–1457, 2015. doi: 10.1109/TIT.2015.2391243. URL <https://arxiv.org/abs/1310.7780>.
- R. Tyrrell Rockafellar. *Convex Analysis*. Princeton University Press, 1970.
- Joel A. Tropp. An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning*, 8(1–2):1–230, 2015. doi: 10.1561/22000000048. URL <https://arxiv.org/abs/1501.01571>.
- Alice Barbara Tumpach and Gabriel Larotonda. Totally geodesic submanifolds in the manifold SPD of symmetric positive-definite real matrices. *Information Geometry*, 7(S2):913–942, 2024. doi: 10.1007/s41884-024-00146-z. URL <https://doi.org/10.1007/s41884-024-00146-z>.
- Lieven Vandenbergh and Stephen Boyd. Semidefinite programming. *SIAM Review*, 38(1):49–95, 1996. doi: 10.1137/1038003.
- Bart Vandereycken, P.-A. Absil, and Stefan Vandewalle. A riemannian geometry with complete geodesics for the set of positive semidefinite matrices of fixed rank. *IMA Journal of Numerical Analysis*, 33(2):481–514, 2013. doi: 10.1093/imanum/drs006. URL <https://doi.org/10.1093/imanum/drs006>.
- Yi Yu, Tengyao Wang, and Richard J. Samworth. A useful variant of the davis–kahan theorem for statisticians. *Biometrika*, 102(2):315–323, 2015. doi: 10.1093/biomet/asv008. URL <https://arxiv.org/abs/1405.0680>.

A Proofs for Metric and Split-Hadamard Reduction

Proof of Theorem 3.2. Assume (A). If $z \in X$ and $r = d_X(x, z)$, ball surjectivity implies $d_Y(q(x), q(z)) \leq d_X(x, z)$; hence q is 1-Lipschitz. For fixed y , set $\delta = d_Y(q(x), y)$. Since $y \in \overline{B}_Y(q(x), \delta)$, submetry gives $z \in \overline{B}_X(x, \delta)$ with $q(z) = y$. Contraction yields

$$\delta \leq d_X(x, z) \leq \delta,$$

proving (B), including attainment.

Assume (B). Taking $y = q(z)$ gives

$$d_Y(q(x), q(z)) = \min_{q(w)=q(z)} d_X(x, w) \leq d_X(x, z),$$

so q is 1-Lipschitz and maps the ambient ball into the visible ball. If y lies in the visible ball, an attaining fiber minimizer lies in the ambient ball. Thus (B) implies (A).

Assume (B), fix z , and put $y = q(z)$. Then $d_X(x, z) \geq d_Y(q(x), y)$, so monotonicity of ρ gives the lower bound from the left side of (6) to the right. An attaining fiber minimizer realizes equality for every y , proving the reverse inequality and hence (C).

Finally assume (C). For $r \geq 0$ and fixed y , use

$$\rho_r(t) = \begin{cases} 0, & t \leq r, \\ +\infty, & t > r, \end{cases} \quad \phi_y(u) = \begin{cases} 0, & u = y, \\ +\infty, & u \neq y. \end{cases}$$

The reduced infimum is zero exactly when $y \in \overline{B}_Y(q(x), r)$, whereas the ambient infimum is zero exactly when a point of $q^{-1}(y)$ lies in $\overline{B}_X(x, r)$. Equality for all y, r is precisely the submetry identity. Thus (C) implies (A). $\square$

Proof of Theorem 3.3. Assume (A). If $q(z) = y$, contraction gives

$$d_X(x, z) \geq d_Y(q(x), y).$$

Because $x = s_x(q(x))$ and s_x is isometric,

$$d_X(x, s_x(y)) = d_Y(q(x), y),$$

which proves (B).

Assume (B), fix y_1, y_2 , and put $x_1 = s_x(y_1)$. Coherence gives $s_{x_1} = s_x$. Applying (B) with anchor x_1 and target y_2 ,

$$d_X(s_x(y_1), s_x(y_2)) = d_Y(q(x_1), y_2) = d_Y(y_1, y_2),$$

so (B) implies (A).

Under (B), every z with $q(z) = y$ has

$$\rho(d_X(x, z)) + \phi(y) \geq \rho(d_Y(q(x), y)) + \phi(y),$$

and $z = s_x(y)$ realizes equality. Taking infima proves universal reduction and canonical minimizer lifting, so (B) implies (C).

Assume (C), choose $\rho(t) = t$, and let ϕ be the indicator of a singleton $\{y\}$. Universal reduction identifies the fiber infimum with $d_Y(q(x), y)$, while the canonical-minimizer clause says that $s_x(y)$ attains it. Hence (C) implies (B).

The isometric-section property gives the reverse ball inclusion exactly as in the proof of Theorem 3.2; hence q is a submetry. For the proximal identity, apply universal reduction with $\rho(t) = t^2/(2\tau)$. A proper lower-semicontinuous geodesically convex function plus squared distance has a unique minimizer in complete CAT(0) space. The section lifts it with equal value. Coherence then gives exact iteration by induction. $\square$

B Proof of the Split-Hadamard Theorem

Lemma B.1 (Complete local isometries over a Hadamard base). *Let $F : L \rightarrow B$ be a local Riemannian isometry from a connected complete Riemannian manifold to a Hadamard manifold. Then F is a global Riemannian isometry.*

Proof. Fix $p \in L$. Any geodesic $\gamma : [0, 1] \rightarrow B$ beginning at $F(p)$ has a unique local lift beginning at p , obtained from a local inverse of F . A lifted segment has the same speed as γ . If its maximal interval ended at $T \leq 1$, then for $s, t \uparrow T$,

$$d_L(\tilde{\gamma}_s, \tilde{\gamma}_t) \leq \text{Len}(\tilde{\gamma}|_{[s, t]}) = \text{Len}(\gamma|_{[s, t]}) \rightarrow 0.$$

Completeness gives a limit point in L , and a local inverse at that point extends the lift past T . Thus every such geodesic lifts over its entire interval. Since every point of the connected complete manifold B is joined to $F(p)$ by a minimizing geodesic, F is surjective.

The same continuation argument gives unique lifting for every piecewise smooth path and for homotopies through such paths; equivalently, a complete local isometry is a covering map. The Hadamard manifold B is simply connected, so its connected covering L has one sheet. Hence F is a diffeomorphism. A bijective local Riemannian isometry preserves lengths in both directions and is therefore a global Riemannian isometry. $\square$

Proof of Theorem 4.2. Because L_x is a horizontal integral leaf, $Dq|_{TL_x}$ is an isometry at every point, so $q|_{L_x}$ is a local Riemannian isometry. The leaf is connected and complete by definition, and the Hadamard base is simply connected. Applying Theorem B.1 shows directly that $q|_{L_x} : L_x \rightarrow B$ is a global Riemannian isometry.

The complete totally geodesic leaf L_x is geodesically convex in M . Indeed, Hopf–Rinow gives an intrinsic minimizing geodesic between any two points of the leaf. Total geodesy makes it an ambient geodesic, and the Hadamard property makes that ambient geodesic the unique globally minimizing one. In particular, intrinsic

and ambient distances agree on L_x . Any horizontal section through x lies in the same maximal integral leaf and equals the inverse s_x of this global isometry.

For every piecewise smooth ambient curve η , Riemannian-submersion contraction gives

$$\text{Len}(q \circ \eta) \leq \text{Len}(\eta),$$

and hence $d_B(q(y), q(z)) \leq d_M(y, z)$. If $q(z) = b$,

$$d_M(x, z) \geq d_B(q(x), b),$$

while $s_x(b)$ attains equality. Suppose another z also attains it, and let η be the ambient minimizing geodesic from x to z . Every inequality in

$$d_B(q(x), b) \leq \text{Len}(q \circ \eta) \leq \text{Len}(\eta) = d_M(x, z)$$

is an equality. The orthogonal decomposition $\dot{\eta} = \dot{\eta}^{\mathcal{H}} + \dot{\eta}^{\mathcal{V}}$ then forces $\dot{\eta}^{\mathcal{V}} = 0$ everywhere. The geodesic stays in L_x and ends at $s_x(b)$, proving uniqueness. The ball identity and radial reduction now follow from Theorem 3.2.

Proximal commutation follows from radial reduction with squared distance. The reduced objective has a unique minimizer on the Hadamard base. Equality in the reduction first forces the compression of any full minimizer to equal that visible proximal point and then forces the full point to be its unique shortest lift from x . Thus the full proximal point is unique even though no geodesic-convexity assertion for $\phi \circ q$ on M is needed. Finally, for $v \in T_z M$,

$$d(\phi \circ q)_z[v] = h_{q(z)}(\text{grad}_B \phi(q(z)), Dq(z)[v]).$$

This vanishes on $\mathcal{V}_z$; on $\mathcal{H}_z$, Dq is an isometry. Riesz representation gives (13). Since Ds_x is the inverse horizontal isometry, the chain rule shows that $z = s_x \circ y$ solves the lifted equation on every interval on which the visible solution y exists. Local Lipschitz continuity gives uniqueness of maximal solutions by the standard ODE theorem. $\square$

Proof of Theorem 4.3. The sections are isometric. If $z = s_x(y)$, then z and x lie on the same maximal horizontal leaf, so both s_z and s_x invert the restriction of q to that leaf. Thus $s_z = s_x$, proving coherence. The unique shortest-lift statement gives rigidity. $\square$

C Proofs for SPD Compression and Completion

Proof of Theorem 5.1. Choose a direct-sum decomposition $E = W \oplus W'$ with $W' \neq \{0\}$. For $t > 0$, let T_t act as the identity on W and as multiplication by t on W' . If a selected positive form P were invariant under every such map, then for nonzero $n \in W'$,

$$P(n, n) = P(T_t n, T_t n) = t^2 P(n, n) \quad \forall t > 0,$$

contradicting $P(n, n) > 0$. Hence no invariant selector exists without a reference or equivalent off-visible structure. $\square$

Proof of Theorem 5.2. Fix $P \in \mathbb{S}_{++}^d$, write $R = A^\top P A$, and let $H = H^\top$ be a tangent at P . Define

$$C = P^{1/2} A R^{-1/2}, \quad X = P^{-1/2} H P^{-1/2}.$$

Then $C^\top C = I_m$ and

$$R^{-1/2} A^\top H A R^{-1/2} = C^\top X C.$$

Complete C to an orthogonal matrix $[C, N]$. The exact block identity is

$$\|X\|_F^2 = \|C^\top X C\|_F^2 + 2 \|N^\top X C\|_F^2 + \|N^\top X N\|_F^2. \quad (61)$$

Therefore

$$\|D\pi_A(P)[H]\|_{R, \text{AI}} = \|C^\top X C\|_F \leq \|X\|_F = \|H\|_{P, \text{AI}}.$$

Integrating along curves proves global contraction (17).

Congruence by $P_0^{-1/2}$ in the ambient cone and by $R_0^{-1/2}$ in the visible cone reduces the completion problem to

$$\min\{d_{\text{AI}}(I, X) : U_0^\top X U_0 = \tilde{R}\}, \quad \tilde{R} = R_0^{-1/2} R R_0^{-1/2}.$$

The candidate

$$X_\star = I + U_0(\tilde{R} - I)U_0^\top$$

is positive definite, has compression $\tilde{R}$, and satisfies

$$d_{\text{AI}}(I, X_\star) = \left\| \log \tilde{R} \right\|_F = d_{\text{AI}}(I, \tilde{R}).$$

Contraction proves it is a minimizer.

For uniqueness, suppose feasible X attains equality and write $L = \log X$. The compressed curve $Y_t = U_0^\top \exp(tL)U_0$ joins I to $\tilde{R}$. Hence

$$d_{\text{AI}}(I, \tilde{R}) \leq \text{Len}(Y) \leq \|L\|_F = d_{\text{AI}}(I, X) = d_{\text{AI}}(I, \tilde{R}).$$

Both length inequalities are therefore equalities. The nonnegative continuous difference between the constant ambient speed and the compressed speed has zero integral, so it vanishes for every t , in particular at $t = 0$. Equality in (61) there forces $N^\top L U_0 = 0$ and $N^\top L N = 0$. Thus $L = U_0 L_0 U_0^\top$, and the endpoint constraint gives $\exp(L_0) = \tilde{R}$. Therefore $X = X_\star$. Undoing congruence gives (16) and (19).

For $R, S \in \mathbb{S}_{++}^m$, the normalized lifts are $\tilde{R} \oplus I$ and $\tilde{S} \oplus I$. Their ambient AIRM distance is the visible AIRM distance, and their geodesic keeps the identity block fixed. Thus the section is isometric and totally geodesic. The ball identity follows by combining contraction with the equal-distance lift.

At a general P , let $L_P(h) = D\Psi_{P,A}(A^\top P A)[h]$. The section identity and isometry give

$$D\pi_A(P)[L_P(h)] = h, \quad \|L_P(h)\|_{P, \text{AI}} = \|h\|_{A^\top P A, \text{AI}}.$$

If $V \in \ker D\pi_A(P)$, then $L_P(h) + sV$ is another lift of h . Infinitesimal contraction gives $\|h\|^2 \leq \|L_P(h) + sV\|^2$ for all s , with equality at zero. Differentiation shows $L_P(h) \perp V$. Surjectivity and dimension counting identify the image of L_P with the full horizontal space. This proves the Riemannian-submersion and horizontal-lift statements.

For an anchor P , write $\mathcal{L}_P = \Psi_{P,A}(\mathbb{S}_{++}^m)$. The sections obey the coherence identity

$$\Psi_{\Psi_{P,A}(S), A}(T) = \Psi_{P,A}(T), \quad S, T \in \mathbb{S}_{++}^m. \quad (62)$$

Indeed, congruence and visible-coordinate covariance reduce the identity to $P = I$ and an orthonormal channel U . Then $Q = \Psi_{I,U}(S) = I + U(S - I)U^\top$, so $QU = US$, and the completion formula gives

$$\Psi_{Q,U}(T) = Q + QUS^{-1}(T - S)S^{-1}U^\top Q = I + U(T - I)U^\top = \Psi_{I,U}(T).$$

Consequently, if $Q \in \mathcal{L}_P$, then $\mathcal{L}_Q = \mathcal{L}_P$. Differentiating this identity at Q , and using the horizontal-lift identification above, yields

$$T_Q \mathcal{L}_P = \text{range } D\Psi_{Q,A}(A^\top Q A) = \mathcal{H}_Q \quad \text{for every } Q \in \mathcal{L}_P.$$

Thus through every ambient point there is a connected embedded integral manifold of the full horizontal distribution. The distribution is therefore integrable. Any horizontal curve beginning at P remains in $\mathcal{L}_P$ by local uniqueness of integral manifolds; hence every connected horizontal integral manifold through P is contained in $\mathcal{L}_P$. Therefore $\mathcal{L}_P$ is the maximal horizontal leaf. The isometry and total-geodesy argument

above also shows that it is complete. The maximal horizontal leaves are thus complete and totally geodesic, proving that π_A is split-Hadamard.

For the minimax statement, if $B = 0$, the bounded fiber is the singleton $\{X_0\}$, and the radius and unique minimax center are immediate. Suppose $B > 0$. Work in the normalized splitting and choose symmetric $H \in \mathbb{R}^{(d-m) \times (d-m)}$ with $\|H\|_F = 1$. The curve

$$X_s = \tilde{R} \oplus \exp(sH)$$

stays in the visible fiber and satisfies $d_{\text{AI}}(X_0, X_s) = |s|$. This proves infinite diameter. The two points X_{-B}, X_B lie in the bounded fiber and are distance $2B$ apart, so every center has worst-case distance at least B . The canonical point X_0 has worst-case distance at most B . If another point attained B , equality in the triangle inequality between the endpoints would make it their midpoint. AIRM is uniquely geodesic, so the midpoint is uniquely X_0 .

Finally, for any admissible (P_0, A) , every P with $R = A^\top P A$ obeys

$$d_{\text{AI}}(P_0, P) \geq d_{\text{AI}}(A^\top P_0 A, R),$$

and the canonical lift realizes equality. Monotonicity of ρ proves both inequalities in (22). The same construction lifts every reduced minimizer. Under strict monotonicity, replacing a noncanonical full point by the strictly nearer canonical point lowers the objective, proving the final correspondence. $\square$

Proof of Theorem 5.3. For $\alpha = Ac$,

$$A^\top u_P = -A^\top P A c = -Rc,$$

so all feasible updates have the same quotient class.

For the converse, use a P_0 -orthogonal splitting of the cotangent space into the visible span and its complement, and choose coordinates in which α has visible coordinate $c_0 \neq 0$. A feasible completion has blocks

$$P = \begin{pmatrix} R & B \\ B^\top & C \end{pmatrix}.$$

The invisible component of $-P\alpha$ is $-B^\top c_0$. Given the coordinate vector z of any desired invisible drift, take

$$B = -\frac{c_0 z^\top}{c_0^\top c_0}.$$

Then $-B^\top c_0 = z$. Choosing

$$C = B^\top R^{-1} B + tI, \quad t > 0,$$

makes the Schur complement positive, hence $P \succ 0$. This realizes every invisible drift. The final claim follows from the block-diagonal form and uniqueness of the canonical completion. $\square$

Proof of Theorem 5.5. Normalize to $P_0 = I$ and an orthogonal frame adapted to U_0 . Every feasible covariance has blocks

$$X = \begin{pmatrix} \tilde{R} & C \\ C^\top & D \end{pmatrix}, \quad S = D - C^\top \tilde{R}^{-1} C \succ 0.$$

Up to constants independent of (C, S) , twice the forward Gaussian KL is

$$\text{tr } X - \log \det X = \text{tr } \tilde{R} - \log \det \tilde{R} + \text{tr } S - \log \det S + \text{tr}(C^\top \tilde{R}^{-1} C).$$

The last term is uniquely minimized at $C = 0$. The scalar inequality $s - \log s \geq 1$ applied to the eigenvalues of S shows that the middle term is uniquely minimized at $S = I$. Hence $X = \tilde{R} \oplus I$, the normalized canonical completion. Congruence returns $\Psi_{P_0, A}(R)$. $\square$

Proof of Theorem 5.6. For any P , put $R = A^\top P A$. Canonical completion gives

$$d_{\text{AI}}(P_0, P) \geq d_{\text{AI}}(R_0, R),$$

with equality exactly at $P = \Psi_{P_0, A}(R)$. The full problem therefore reduces exactly to

$$\min_{R \in \mathbb{S}_{++}^m} (1 - \tau) d_{\text{AI}}(R_0, R)^2 + \tau d_{\text{AI}}(R, \hat{R})^2.$$

Let $x = d_{\text{AI}}(R_0, R)$, $y = d_{\text{AI}}(R, \hat{R})$, and $D = d_{\text{AI}}(R_0, \hat{R})$. The triangle inequality gives $x + y \geq D$, while

$$(1 - \tau)x^2 + \tau y^2 = \tau(1 - \tau)(x + y)^2 + ((1 - \tau)x - \tau y)^2.$$

The lower bound $\tau(1 - \tau)D^2$ is attained only when R lies on the unique AIRM geodesic from R_0 to $\hat{R}$ with $x = \tau D$, $y = (1 - \tau)D$. Thus $R = R_0 \#_\tau \hat{R}$, and canonical completion proves (27) and the distance identities.

For robustness, geodesic convexity of distance in a Hadamard manifold gives

$$d_{\text{AI}}(R_0 \#_\tau R_1, R_0 \#_\tau R_2) \leq \tau d_{\text{AI}}(R_1, R_2).$$

The completion section is isometric, so the same bound holds for the full solutions. $\square$

D Proofs for Rank-Stratified Geometry

Proof of Theorem 6.1. Let Fr_m be the manifold of full-column-rank $d \times m$ frames. Its right $GL(m)$ action is free and proper, with quotient $\text{Gr}(m, d)$. The diagonal action on $\text{Fr}_m \times \mathbb{S}_{++}^m$ used in (28) is also free. It is proper by the sequential criterion: if (A_k, R_k) and $(A_k B_k, B_k^\top R_k B_k)$ both converge inside the product, then smooth left inverses of the full-rank A_k show that B_k converges; the limit is invertible because the limiting transformed frame has rank m . Consequently the quotient is a smooth manifold. Equivalently, it is the smooth open cone bundle

$$\{(W, g) : W \in \text{Gr}(m, d), g \text{ is a positive-definite bilinear form on } W\}$$

over the Grassmannian. The identification sends $[A, R]$ to $W = \text{range}(A)$ and the form defined by $g(Ax, Ay) = x^\top R y$. This formula is independent of the representative. The regular descriptor space is the open subbundle on which $g - P_0|_W$ is nondegenerate.

Gauge invariance follows because the replacement $(A, R) \mapsto (AB, B^\top R B)$ leaves $P_0 A R_0^{-1} (R - R_0) R_0^{-1} A^\top P_0$ unchanged. Moreover,

$$\Psi_{P_0, A}(R) - P_0 = P_0 A R_0^{-1} (R - R_0) R_0^{-1} A^\top P_0. \quad (63)$$

If $R - R_0$ is invertible, this difference has rank m .

Conversely, let $P \in \mathfrak{S}_m(P_0)$, put $H = P - P_0$, and define $\mathcal{W} = P_0^{-1} \text{range}(H)$. Choose full-column-rank A spanning $\mathcal{W}$. Then $P_0 A$ spans $\text{range}(H)$. Since H is symmetric of rank m , there is an invertible symmetric C such that

$$H = P_0 A C A^\top P_0.$$

For example, with $B = P_0 A$, take

$$C = (B^\top B)^{-1} B^\top H B (B^\top B)^{-1}.$$

The identity follows because the range and co-range of H both equal $\text{range}(B)$. Setting $R = A^\top P A$ and $R_0 = A^\top P_0 A$ gives

$$R - R_0 = A^\top H A = R_0 C R_0,$$

which is invertible, and (63) reconstructs P .

The subspace $\mathcal{W}$ is intrinsic. Two frames for it differ by a unique $B \in GL(m)$, and their compressed matrices differ by congruence. This proves bijectivity. It remains to verify the smooth structures rather than only the

set-theoretic inverse. After whitening P_0 , the bundle description above identifies a regular descriptor with a pair (W, T) , where T is a positive-definite self-adjoint endomorphism of W and $T - I_W$ is invertible. In a local orthonormal frame $U(W)$, the map is

$$(W, T) \mapsto I + U(W)([T]_{U(W)} - I)U(W)^\top,$$

so it is smooth and has rank- m displacement. Conversely, on the fixed-rank symmetric displacement manifold, the nonzero spectrum of $H = P - P_0$ remains locally separated from zero. The range projector Π_H is therefore smooth by spectral functional calculus. A local smooth frame of $\text{range}(\Pi_H)$, together with the restriction of P to that range, gives exactly the local inverse (W, T) . Transition between two such frames is smooth congruence, which is the quotient transition already built into the form bundle. Fixed inertia classes of H are open-and-closed components of the fixed-rank symmetric manifold; the same argument applies on every component, so no signature branch is omitted. Hence Γ_{m, P_0} and its inverse are smooth, and the map is a diffeomorphism.

Every SPD matrix lies in exactly one rank stratum. A limit of rank- m symmetric differences has rank at most m . Conversely, if $\text{rank}(P - P_0) = r < m$, whiten by P_0 , choose $m - r$ orthonormal kernel directions of the whitened difference, and add arbitrarily small nonzero eigenvalues in those directions. The perturbation stays SPD for sufficiently small magnitudes, has rank m , and converges to P . This proves (34). $\square$

Proof of Theorem 6.2. Under $A^\top P_0 A = I$, put $U = P_0^{1/2} A$. Whitening sends the base point and the differential represented by $\xi_i = (H_i, K_i)$ to

$$\Gamma \longleftrightarrow \begin{pmatrix} R & 0 \\ 0 & I \end{pmatrix}, \quad D\tilde{\Gamma}[\xi_i] \longleftrightarrow \begin{pmatrix} H_i & (R - I)K_i^\top \\ K_i(R - I) & 0 \end{pmatrix}.$$

Using $\langle X, Y \rangle_{P, \text{AI}} = \text{tr}(P^{-1} X P^{-1} Y)$, direct block multiplication for directions i, j gives

$$\text{tr}(R^{-1} H_i R^{-1} H_j) + 2 \text{tr}(K_i(R - I) R^{-1} (R - I) K_j^\top),$$

which is (35). Setting $i = j$ proves (37). Factoring its two blocks gives (38). The pair matrix is therefore a Gram matrix, which proves its positive semidefiniteness, the rank identity (39), and the orthogonality criterion (40).

The differential vanishes exactly when $H = 0$ and $K(R - I) = 0$, proving (41). If $m < d$ and $R - I$ is singular, a nonzero row in its nullspace can be placed in K , producing a nonzero kernel direction; if $R - I$ is invertible, the kernel condition gives $K = 0$. Thus for $m < d$ the pullback is positive definite exactly on the regular domain of Theorem 6.1. If $m = d$, the horizontal Grassmann block has dimension zero, so K is absent and the visible term is positive definite for every $R \in \mathbb{S}_{++}^d$. In this case the base Grassmannian is a point and, after whitening, the total descriptor is simply a positive form on the full space; the completion map returns that same form. It is therefore a global diffeomorphism onto $\mathbb{S}_{++}^d$.

The eigenvalues of $\Xi(R)$ are $(\lambda_i(R) - 1)^2 / \lambda_i(R)$. Under the spectral assumptions,

$$\lambda_{\min}(\Xi(R)) \geq \frac{\zeta^2}{b}, \quad \lambda_{\max}(\Xi(R)) \leq \max_{x \in [a, b]} \frac{(x - 1)^2}{x}.$$

Substitution into the exact line element proves (42). $\square$

Proof of Theorem 6.3. The shortest-completion identity (19) gives

$$\min_{A^\top P A = R} \frac{1}{2} d_{\text{AI}}(P_0, P)^2 = \frac{1}{2} d_{\text{AI}}(A^\top P_0 A, R)^2.$$

Under the normalization $A^\top P_0 A = I$, this is (43) and is independent of the horizontal subspace coordinate. Derivatives with respect to that coordinate at fixed R consequently vanish. The minimizer is the explicit section $P = \Psi_{P_0, A}(R)$, so evaluating it requires no further vertical stationarity equation. Differentiating that section instead measures the motion of the minimizing ambient matrix, whose AIRM inner product is (35); it is not the Hessian of the reduced value. $\square$

Frozen-relaxed decomposition in the two-dimensional example. The cancellation mentioned in Theorem 6.4 can be made explicit in fixed ambient coordinates. Write

$$u(\theta) = (\cos \theta, \sin \theta)^\top, \quad P = \begin{pmatrix} a & c \\ c & b \end{pmatrix},$$

and eliminate the constraint $u(\theta)^\top P u(\theta) = r$ as

$$a = \frac{r - 2c \sin \theta \cos \theta - b \sin^2 \theta}{\cos^2 \theta}.$$

For $F(b, c, \theta) = \frac{1}{2} \|\log P\|_F^2$, the hidden minimizer at $\theta = 0$ is $(b, c) = (1, 0)$. Direct differentiation there gives

$$F_{\theta\theta} = \frac{2(r-1)\log r}{r}, \quad F_{\theta c} = -\frac{2\log r}{r}, \quad F_{cc} = \frac{2\log r}{r(r-1)},$$

with $F_{\theta b} = F_{bc} = 0$ and $F_{bb} = 1$. The hidden Hessian is positive definite because $\log r / (r-1) > 0$. Consequently the relaxed second derivative is

$$F_{\theta\theta} - F_{\theta z} F_{zz}^{-1} F_{z\theta} = \frac{2(r-1)\log r}{r} - \frac{2(r-1)\log r}{r} = 0, \quad z = (b, c).$$

Thus the frozen direct term and the nonzero Schur term cancel, consistently with the constant reduced value. The response $c'(0) = -F_{cc}^{-1} F_{c\theta} = r-1$ is the off-diagonal derivative of the canonical completion in Theorem 6.4.

Proof of Theorem 6.5. By definition, $P_t = G_t Q_t G_t^\top$. Differentiating and multiplying by G_t^{-1} and $G_t^{-\top}$ gives

$$G_t^{-1} \dot{P}_t G_t^{-\top} = \dot{Q}_t + \Omega_t Q_t + Q_t \Omega_t^\top = \mathfrak{D}_t Q_t.$$

Congruence invariance of AIRM yields (44).

If $G'_t = G_t O_t$, then

$$Q'_t = O_t^\top Q_t O_t, \quad \Omega'_t = O_t^\top \Omega_t O_t + O_t^\top \dot{O}_t.$$

Differentiating Q'_t and substituting these identities cancels all terms containing $O_t^\top \dot{O}_t$, leaving

$$\mathfrak{D}'_t Q'_t = O_t^\top (\mathfrak{D}_t Q_t) O_t.$$

Orthogonal congruence preserves the displayed Frobenius norm. $\square$

Proof of Theorem 6.6. After whitening, the nonzero singular values of

$$P_t - I = U_t (R_t - I) U_t^\top$$

are exactly those of $R_t - I$. Singular values vary continuously with the matrix. If the limit has rank at most $m-r$, at least r of the first m singular values converge to zero. The same therefore holds for $R_t - I$. $\square$

E Proofs for Structured Expressivity

Lemma E.1 (Linear images of relative interiors). *Let C be a nonempty convex set in a finite-dimensional vector space and let L be linear. Then*

$$L(\text{ri } C) = \text{ri } L(C). \quad (64)$$

Proof. Let $x \in \text{ri } C$, set $y = Lx$, and choose any $z = Lx_z \in L(C)$. The relative-interior extension property gives $\eta > 0$ with $x + \eta(x - x_z) \in C$. Applying L gives $y + \eta(y - z) \in L(C)$, proving $y \in \text{ri } L(C)$.

Conversely, let $y \in \text{ri } L(C)$, choose $x_0 \in \text{ri } C$, and set $z = Lx_0$. The extension property in $L(C)$ gives $\eta > 0$ and $x_1 \in C$ satisfying $Lx_1 = y + \eta(y - z)$. Then

$$x = \frac{x_1 + \eta x_0}{1 + \eta}$$

belongs to $\text{ri } C$ and satisfies $Lx = y$. $\square$

Proof of Theorem 7.1. Let $L : \mathbb{R}^d \rightarrow \mathbb{S}^m$ be $L(p) = \sum_j p_j G_j$. Since $\text{ri } \mathbb{R}_+^d = \mathbb{R}_{++}^d$, Theorem E.1 gives

$$L(\mathbb{R}_{++}^d) = \text{ri } L(\mathbb{R}_+^d) = \text{ri } \mathcal{K}_{\text{diag}}.$$

This proves the feasibility criterion. With $y_j = \log p_j$, the objective is $\|y - y^0\|_2$ and the constraint is $\sum_j e^{y_j} G_j = R$. A minimizing sequence is bounded in y , hence has a convergent subsequence whose limit remains feasible. Thus the infimum is attained. Linear independence of the G_j gives uniqueness of the coefficients.

The cone $\mathcal{K}_{\text{diag}}$ is finitely generated and closed. If R lies outside it, closed-cone separation gives symmetric Y with $\text{tr}(YR) < 0$ and $\text{tr}(YG_j) \geq 0$ for all j . If R lies on its relative boundary, apply the supporting-hyperplane theorem in $S = \text{span } \mathcal{K}_{\text{diag}}$. It gives a linear functional on S , nonzero on S , that vanishes at R and is nonnegative on the cone. Represent it by a symmetric Y . Then $\text{tr}(YR) = 0$ and $\text{tr}(YG_j) \geq 0$ for every j . At least one of these generator pairings is strictly positive: otherwise the functional would vanish on the span of all generators, namely on S . This proves the stated facial certificate and excludes ambient annihilators of the cone. $\square$

Proof of Theorem 7.3. Let $C = \prod_b \mathbb{S}_+^{(b)}$ and

$$L((P_b)_b) = \sum_b A_b^\top P_b A_b.$$

The relative interior of C is $\prod_b \mathbb{S}_{++}^{(b)}$. Therefore Theorem E.1 gives

$$L\left(\prod_b \mathbb{S}_{++}^{(b)}\right) = \text{ri } L(C) = \text{ri } \mathcal{K}_{\text{block}}.$$

This proves strict block feasibility. If a feasible sequence has bounded objective, the generalized eigenvalues of each $(P_b^{(n)}, P_b^0)$ stay in a common compact interval $[e^{-C_0}, e^{C_0}]$. A subsequence converges to positive definite blocks, and the linear equality is closed, proving attainment.

For each block, the image cone

$$C_b = \{A_b^\top P_b A_b : P_b \succeq 0\}$$

is exactly the cone of PSD matrices whose range is contained in $\text{range}(A_b^\top)$, and is therefore closed. To see that their finite sum is closed, let $X_n = \sum_b X_{b,n}$ converge with $X_{b,n} \in C_b$. Every summand is PSD, so $0 \leq \text{tr}(X_{b,n}) \leq \text{tr}(X_n)$. The summands are consequently bounded; passing to common subsequences gives $X_{b,n} \rightarrow X_b \in C_b$ and $X = \sum_b X_b$. Hence $\mathcal{K}_{\text{block}} = \sum_b C_b$ is closed.

If $R \notin \mathcal{K}_{\text{block}}$, separation from the closed cone gives symmetric Y with $\text{tr}(YR) < 0$ and

$$\text{tr}\left(Y \sum_b A_b^\top P_b A_b\right) \geq 0 \quad \text{for all } P_b \succeq 0.$$

Blockwise duality of the PSD cone makes this equivalent to $A_b Y A_b^\top \succeq 0$ for every b . If R lies on the relative boundary, take a supporting functional that is nonzero on $S = \text{span } \mathcal{K}_{\text{block}}$. It gives the same blockwise condition with $\text{tr}(YR) = 0$. Some $A_b Y A_b^\top$ must be nonzero: if every one vanished, the functional would vanish on every image $A_b^\top P_b A_b$, hence on S . $\square$

F Proofs for Deterministic and Statistical Recovery

Proof of Theorem 8.1. Set

$$P = \mathfrak{C}(U, R), \quad P_1 = \mathfrak{C}(U, S), \quad Q = \mathfrak{C}(V, S).$$

At fixed U , the completion section is isometric, so $d_{\text{AI}}(P, P_1) = d_{\text{AI}}(R, S)$. Let U_t be the horizontal minimizing Grassmann geodesic from U to V , and keep S fixed in the parallel frame. The exact line element gives

$$\|\dot{Q}_t\|_{Q_t, \text{AI}}^2 = 2 \operatorname{tr}(K_t \Xi(S) K_t^\top) \leq 2\chi(S)^2 \|K_t\|_F^2.$$

Therefore

$$d_{\text{AI}}(P_1, Q) \leq \sqrt{2}\chi(S) \int_0^1 \|K_t\|_F dt = \sqrt{2}\chi(S) d_{\text{Gr}}(W, \widetilde{W}).$$

The triangle inequality gives the claimed bound with $\chi(S)$. Reversing the order, first rotating with R and then changing visible geometry at fixed V , gives the bound with $\chi(R)$. Taking the smaller proves (54). $\square$

Proof of Theorem 8.2. Let $Y_i = R^{-1/2} X_i$. Then

$$\mathbb{E}[Y_i Y_i^\top] = I, \quad \|Y_i\|_2 \leq L,$$

and

$$\widehat{S} = R^{-1/2} \widehat{R} R^{-1/2} = \frac{1}{n} \sum_{i=1}^n Y_i Y_i^\top.$$

The Y_i are i.i.d. For fixed unit u , the variables $Z_i = (u^\top Y_i)^2$ satisfy $0 \leq Z_i \leq L^2$ and $\mathbb{E} Z_i = 1$. Hoeffding's inequality gives

$$\Pr \left\{ \left| \frac{1}{n} \sum_i Z_i - 1 \right| \geq t \right\} \leq 2 \exp \left(-\frac{2nt^2}{L^4} \right).$$

Let $\mathcal{N}$ be a $1/4$ -net of the unit sphere in $\mathbb{R}^m$, with $|\mathcal{N}| \leq 9^m$, and take $t = \varepsilon/4$. A union bound and the sample-size assumption show that, with probability at least $1 - \delta$,

$$|u^\top (\widehat{S} - I) u| \leq \varepsilon/4 \quad \forall u \in \mathcal{N}.$$

For symmetric H , a $1/4$ -net obeys

$$\|H\|_{\text{op}} \leq 2 \sup_{u \in \mathcal{N}} |u^\top H u|.$$

Indeed, if unit x attains the operator norm and $\|x - u\| \leq 1/4$, then $|x^\top H x - u^\top H u| \leq \frac{1}{2} \|H\|_{\text{op}}$. Hence $\|\widehat{S} - I\|_{\text{op}} \leq \varepsilon/2$. For $0 < \varepsilon \leq 1$,

$$e^{-\varepsilon} I \preceq (1 - \varepsilon/2) I \preceq \widehat{S} \preceq (1 + \varepsilon/2) I \preceq e^\varepsilon I.$$

Congruence by $R^{1/2}$ gives the Loewner sandwich. Its generalized log-eigenvalues have magnitude at most ε , so $d_{\text{AI}}(R, \widehat{R}) \leq \sqrt{m} \varepsilon$. Fixed-channel completion is isometric, proving (56). $\square$

Lemma F.1 (Spectral-gap subspace perturbation). *Let $\widehat{\Sigma} = \Sigma + E$, $\|E\|_{\text{op}} \leq \delta < \gamma$, and let V span the leading m -dimensional eigenspace of $\widehat{\Sigma}$. Then*

$$\|U_\perp^\top V\|_{\text{op}} \leq s_\delta = \frac{\delta}{\gamma - \delta}. \quad (65)$$

If $\delta < \gamma/2$, then

$$d_{\text{Gr}}(W, \widehat{W}) \leq \sqrt{m} \arcsin(s_\delta), \quad (66)$$

and after orthogonal Procrustes alignment,

$$\|V - U\|_{\text{op}} \leq \sqrt{2} s_\delta. \quad (67)$$

Proof. In the orthogonal basis $[U, U_\perp]$, write

$$\Sigma = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}, \quad A \succeq \lambda_m I, \quad C \preceq \lambda_{m+1} I.$$

Let $\widehat{\Lambda}$ contain the leading eigenvalues of $\widehat{\Sigma}$. Weyl's inequality gives $\widehat{\Lambda} \succeq (\lambda_m - \delta)I$. With $Y = U_\perp^\top V$, projection of $\widehat{\Sigma}V = V\widehat{\Lambda}$ gives the Sylvester equation

$$Y\widehat{\Lambda} - CY = F, \quad F = U_\perp^\top EV.$$

The spectra are separated by at least $\gamma - \delta$, and

$$Y = \int_0^\infty e^{tC} F e^{-t\widehat{\Lambda}} dt.$$

Thus

$$\|Y\|_{\text{op}} \leq \int_0^\infty e^{-t(\gamma-\delta)} \delta dt = \frac{\delta}{\gamma - \delta}.$$

The singular values of $U_\perp^\top V$ are the sines of the principal angles, proving (66). In principal frames,

$$\|V - U\|_{\text{op}} = \max_i 2 \sin(\theta_i/2) \leq \sqrt{2} \max_i \sin \theta_i \leq \sqrt{2} s_\delta,$$

which proves (67). $\square$

Proof of Theorem 8.3. Align V to U as in Theorem F.1, and conjugate $\widehat{R}$ by the same orthogonal alignment. Denote the aligned matrix by S ; the completed estimator is unchanged. Then

$$\begin{aligned} \|S - R\|_{\text{op}} &\leq \left\| V^\top (\widehat{\Sigma} - \Sigma) V \right\|_{\text{op}} + \|V^\top \Sigma V - U^\top \Sigma U\|_{\text{op}} \\ &\leq \delta + 2\lambda_1 \|V - U\|_{\text{op}} \\ &\leq \delta + 2\sqrt{2}\lambda_1 s_\delta = \varepsilon_{\text{vis}}. \end{aligned}$$

Since $R \succeq \lambda_m I$,

$$\left\| R^{-1/2} (S - R) R^{-1/2} \right\|_{\text{op}} \leq \eta_\delta < 1.$$

Therefore

$$(1 - \eta_\delta)R \preceq S \preceq (1 + \eta_\delta)R,$$

and

$$d_{\text{AI}}(R, S) \leq \sqrt{m}[-\log(1 - \eta_\delta)]. \quad (68)$$

Weyl's inequality places the eigenvalues of S in $[\lambda_m - \delta, \lambda_1 + \delta]$, so $\chi(S) \leq \omega_\delta$. Apply Theorems F.1 and 8.1 and (68):

$$\begin{aligned} d_{\text{AI}}(P, \widehat{P}) &\leq d_{\text{AI}}(R, S) + \sqrt{2}\chi(S)d_{\text{Gr}}(W, \widehat{W}) \\ &\leq \sqrt{m}[-\log(1 - \eta_\delta)] + \sqrt{2}\omega_\delta\sqrt{m}\arcsin(s_\delta). \end{aligned}$$

This is (57). $\square$

Lemma F.2 (Ambient second-moment concentration). *Under the assumptions of Theorem 8.4, with probability at least $1 - \alpha$,*

$$\left\| \widehat{\Sigma} - \Sigma \right\|_{\text{op}} \leq \delta_n(\alpha).$$

Proof. Let $\mathcal{N}$ be a $1/4$ -net of the Euclidean unit sphere with $|\mathcal{N}| \leq 9^d$. For fixed $u \in \mathcal{N}$, the i.i.d. variables $Z_i = (u^\top X_i)^2$ lie in $[0, L^2]$. Hoeffding and a union bound give

$$\Pr \left\{ \sup_{u \in \mathcal{N}} |u^\top (\hat{\Sigma} - \Sigma) u| \geq t \right\} \leq 2 \cdot 9^d \exp \left(-\frac{2nt^2}{L^4} \right).$$

The net estimate used above gives $\left\| \hat{\Sigma} - \Sigma \right\|_{\text{op}} \leq 2t$. Taking $t = \delta_n(\alpha)/2$ makes the failure probability at most α . $\square$

Proof of Theorem 8.4. On the event from Theorem F.2, all assumptions of Theorem 8.3 hold with $\delta = \delta_n(\alpha)$. Apply that theorem. For the explicit sufficient condition, $\delta \leq \delta_\star \leq \gamma/4$ gives

$$s_\delta = \frac{\delta}{\gamma - \delta} \leq \frac{4\delta}{3\gamma}.$$

Consequently

$$\varepsilon_{\text{vis}} \leq \delta \left(1 + \frac{8\sqrt{2}\lambda_1}{3\gamma} \right) = c_\gamma \delta \leq \frac{\lambda_m}{2}, \quad \eta_\delta \leq \frac{1}{2}.$$

Also $\delta \leq \gamma/4 < \gamma/2$ and $\delta \leq \lambda_m/(2c_\gamma) < \lambda_m$, so every hypothesis of Theorem 8.3 is satisfied. $\square$